sky130_rhbd_tt_1P8_25C.ccs Library

Cell Groups
AND2X1
AND3X1
AO3X1
AOA4X1
AOAI4X1
AOI3X1
BUFX1
DFFQNX1
DFFQX1
DFFRNQNX1
DFFRNQX1
DFFRNX1
DFFSNQNX1
DFFSNQX1
DFFSNRNQNX1
DFFSNRNQX1
DFFSNRNX1
DFFSNX1
DFFX1
DLATCHN
DLATCH
FA
HA

INVX1
INV
MUX2X1
NAND2X1
NAND3X1
NOR2X1
NOR3X1
OR2X1
OR3X1
TIEHI
TIELO
TMRDFFQNX1
TMRDFFQX1
TMRDFFRNQNX1
TMRDFFRNQX1
TMRDFFSNQNX1
TMRDFFSNQX1
TMRDFFSNRNQNX1
TMRDFFSNRNQX1
VOTER3X1
VOTERN3X1
XNOR2X1
XOR2X1

AND2X1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT		OUTPUT
A	B	Y
0	x	0
1	0	0
1	1	1

Footprint

Cell Name	Area
AND2X1	50.67280

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	A	B	Y
AND2X1	0.01053	0.01062	5.78592

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
AND2X1	0.00000	7.37293	13.28960

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AND2X1	A->Y (RR)	0.06039	0.37906	7.29315
	B->Y (RR)	0.05367	0.37982	7.30317

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AND2X1	A->Y (FF)	0.05120	0.35220	6.14424
	B->Y (FF)	0.04774	0.34423	5.96601

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
AND2X1	A	0.00000	0.00000	0.00000
	A	8.25035	8.25909	8.55443
	B	0.00000	0.00000	0.00000
	B	8.25152	8.26165	8.56390

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
AND2X1	A	0.00000	0.00000	0.00000
	A	2.39127	2.40576	2.75402
	B	0.00000	0.00000	0.00000
	B	2.38391	2.39316	2.66566

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND2X1	(!B * !Y)	0.00000	0.00000	0.00000
	(!B * !Y)	2.55829	2.55792	2.55772

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND2X1	(!B * !Y)	0.00000	0.00000	0.00000
	(!B * !Y)	3.15060	3.14973	3.14975

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND2X1	(!A * !Y)	0.00000	0.00000	0.00000
	(!A * !Y)	2.56007	2.55952	2.55962

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND2X1	(!A * !Y)	0.00000	0.00000	0.00000
	(!A * !Y)	3.15109	3.15097	3.15141

AND3X1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT			OUTPUT
A	B	C	Y
0	x	x	0
1	0	x	0
1	1	0	0
1	1	1	1

Footprint

Cell Name	Area
AND3X1	62.15760

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	A	B	C	Y
AND3X1	0.01056	0.01034	0.01054	5.79380

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
AND3X1	0.00000	6.97633	20.76290

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AND3X1	A->Y (RR)	0.08659	0.41334	7.47699
	B->Y (RR)	0.07975	0.41821	7.50064
	C->Y (RR)	0.07103	0.42059	7.56135

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AND3X1	A->Y (FF)	0.05895	0.37074	6.18891
	B->Y (FF)	0.05601	0.36259	5.91755
	C->Y (FF)	0.05114	0.35440	5.89424

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
AND3X1	A	0.00000	0.00000	0.00000
	A	13.03860	13.04420	13.31230
	B	0.00000	0.00000	0.00000
	B	13.03910	13.04500	13.31150
	C	0.00000	0.00000	0.00000
	C	13.04000	13.04690	13.31700

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
AND3X1	A	0.00000	0.00000	0.00000
	A	1.66239	1.67514	2.01460
	B	0.00000	0.00000	0.00000
	B	1.65420	1.66434	1.94045
	C	0.00000	0.00000	0.00000
	C	1.64628	1.65414	1.87982

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND3X1	(B * !C * !Y)	0.00000	0.00000	0.00000
	(B * !C * !Y)	2.22186	2.22148	2.22122
	(!B * C * !Y)	0.00000	0.00000	0.00000
	(!B * C * !Y)	2.22478	2.22447	2.22424
	(!B * !C * !Y)	0.00000	0.00000	0.00000
	(!B * !C * !Y)	2.22075	2.22000	2.22057

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND3X1	(B * !C * !Y)	0.00000	0.00000	0.00000
	(B * !C * !Y)	2.78081	2.77974	2.77976
	(!B * C * !Y)	0.00000	0.00000	0.00000
	(!B * C * !Y)	2.78219	2.78133	2.78121
	(!B * !C * !Y)	0.00000	0.00000	0.00000
	(!B * !C * !Y)	2.77901	2.77862	2.77875

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND3X1	(A * !C * !Y)	0.00000	0.00000	0.00000
	(A * !C * !Y)	2.22199	2.22149	2.22163
	(!A * C * !Y)	0.00000	0.00000	0.00000
	(!A * C * !Y)	2.22738	2.22710	2.22663
	(!A * !C * !Y)	0.00000	0.00000	0.00000
	(!A * !C * !Y)	2.22090	2.21968	2.22061

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND3X1	(A * !C * !Y)	0.00000	0.00000	0.00000
	(A * !C * !Y)	2.78025	2.77929	2.77927
	(!A * C * !Y)	0.00000	0.00000	0.00000
	(!A * C * !Y)	2.78221	2.78205	2.78212
	(!A * !C * !Y)	0.00000	0.00000	0.00000
	(!A * !C * !Y)	2.78035	2.77989	2.78037

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND3X1	(!B * !Y)	0.00000	0.00000	0.00000
	(!B * !Y)	2.22438	2.22356	2.22389
	(!A * B * !Y)	0.00000	0.00000	0.00000
	(!A * B * !Y)	2.22821	2.22698	2.22757

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND3X1	(!B * !Y)	0.00000	0.00000	0.00000
	(!B * !Y)	2.78079	2.78107	2.78092
	(!A * B * !Y)	0.00000	0.00000	0.00000
	(!A * B * !Y)	2.78017	2.77989	2.78003

AO3X1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT			OUTPUT
A	B	C	Y
0	x	x	1
1	0	x	1
1	1	0	0
1	1	1	1

Footprint

Cell Name	Area
AO3X1	76.51360

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	A	B	C	Y
AO3X1	0.01015	0.01038	0.01039	5.46339

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
AO3X1	0.00000	5.28276	19.23110

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AO3X1	A->Y (FR)	0.09498	0.46398	7.45046
	B->Y (FR)	0.08975	0.44287	7.11280
	C->Y (RR)	0.04142	0.34573	6.60790

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AO3X1	A->Y (RF)	0.13590	0.39922	4.93534
	B->Y (RF)	0.12787	0.39427	4.83945
	C->Y (FF)	0.06978	0.38945	6.06601

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
AO3X1	A	0.00000	0.00000	0.00000
	A	1.49595	1.50793	1.76975
	B	0.00000	0.00000	0.00000
	B	1.48627	1.49619	1.72388
	C	0.00000	0.00000	0.00000
	C	11.37790	11.38600	11.61050

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
AO3X1	A	0.00000	0.00000	0.00000
	A	10.20580	10.21120	10.39770
	B	0.00000	0.00000	0.00000
	B	10.20720	10.21240	10.41000
	C	0.00000	0.00000	0.00000
	C	11.39380	11.40280	11.62910

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AO3X1	(B * C * Y)	0.00000	0.00000	0.00000
	(B * C * Y)	11.81610	11.82090	12.05330
	(!B * Y)	0.00000	0.00000	0.00000
	(!B * Y)	0.03478	0.03475	0.03473

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AO3X1	(B * C * Y)	0.00000	0.00000	0.00000
	(B * C * Y)	0.06908	0.07840	0.34017
	(!B * Y)	0.00000	0.00000	0.00000
	(!B * Y)	0.05846	0.05789	0.05783

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AO3X1	(A * C * Y)	0.00000	0.00000	0.00000
	(A * C * Y)	11.81640	11.82330	12.06040
	(!A * Y)	0.00000	0.00000	0.00000
	(!A * Y)	0.03746	0.03740	0.03740

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AO3X1	(A * C * Y)	0.00000	0.00000	0.00000
	(A * C * Y)	0.06016	0.06812	0.29064
	(!A * Y)	0.00000	0.00000	0.00000
	(!A * Y)	0.05804	0.05824	0.05819

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AO3X1	(A * !B * Y) + (!A * Y)	0.00000	0.00000	0.00000
	(A * !B * Y) + (!A * Y)	0.02401	0.02398	0.02398

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	**mid**	**last**
AO3X1	**(A * !B * Y) + (!A * Y)**	0.00000	0.00000	0.00000
	(A * !B * Y) + (!A * Y)	1.47366	1.47352	1.47376

AOA4X1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT				OUTPUT
A	B	C	D	Y
0	x	x	x	0
1	0	x	x	0
1	1	x	0	0
1	1	0	1	1
1	1	1	1	0

Footprint

Cell Name	Area
AOA4X1	102.35440

Pin Capacitance Information

Cell Name	Pin Cap(pf)				Max Cap(pf)
	A	B	C	D	Y
AOA4X1	0.01014	0.01034	0.01030	0.01037	5.34524

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
AOA4X1	0.00000	10.36610	25.24330

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AOA4X1	A->Y (RR)	0.19397	0.51161	6.69779
	B->Y (RR)	0.18593	0.51324	6.79922
	C->Y (FR)	0.12648	0.49946	7.61556
	D->Y (RR)	0.05570	0.38309	6.94448

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AOA4X1	A->Y (FF)	0.13779	0.43228	5.28877
	B->Y (FF)	0.13273	0.41780	5.16986
	C->Y (RF)	0.08338	0.31866	4.23334
	D->Y (FF)	0.04885	0.34296	5.58199

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
AOA4X1	A	0.00000	0.00000	0.00000
	A	14.30530	14.30820	14.53220
	B	0.00000	0.00000	0.00000
	B	14.30890	14.31150	14.54430
	C	0.00000	0.00000	0.00000
	C	15.16060	15.16830	15.37330
	D	0.00000	0.00000	0.00000
	D	15.24160	15.24860	15.49130

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
AOA4X1	A	0.00000	0.00000	0.00000
	A	4.88007	4.89061	5.18739
	B	0.00000	0.00000	0.00000
	B	4.87168	4.88019	5.13920
	C	0.00000	0.00000	0.00000
	C	14.49880	14.50360	14.70480
	D	0.00000	0.00000	0.00000
	D	9.30764	9.31653	9.54753

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOA4X1	(B * C * !Y)	0.00000	0.00000	0.00000
	(B * C * !Y)	15.38050	15.38530	15.61250
	(B * !C * !D * !Y)	0.00000	0.00000	0.00000
	(B * !C * !D * !Y)	8.11314	8.11346	8.29790
	(!B * !Y)	0.00000	0.00000	0.00000
	(!B * !Y)	2.75264	2.75235	2.75264

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOA4X1	(B * C * !Y)	0.00000	0.00000	0.00000
	(B * C * !Y)	2.51238	2.52115	2.77697
	(B * !C * !D * !Y)	0.00000	0.00000	0.00000
	(B * !C * !D * !Y)	5.02243	5.03100	5.27043
	(!B * !Y)	0.00000	0.00000	0.00000
	(!B * !Y)	3.06278	3.06214	3.06212

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOA4X1	(A * C * !Y)	0.00000	0.00000	0.00000
	(A * C * !Y)	15.38060	15.38560	15.61730
	(A * !C * !D * !Y)	0.00000	0.00000	0.00000
	(A * !C * !D * !Y)	8.11480	8.11701	8.30555
	(!A * !Y)	0.00000	0.00000	0.00000
	(!A * !Y)	2.75409	2.75414	2.75445

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOA4X1	(A * C * !Y)	0.00000	0.00000	0.00000
	(A * C * !Y)	2.50210	2.51156	2.73632
	(A * !C * !D * !Y)	0.00000	0.00000	0.00000
	(A * !C * !D * !Y)	5.01458	5.02205	5.23704
	(!A * !Y)	0.00000	0.00000	0.00000
	(!A * !Y)	3.06284	3.06282	3.06297

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOA4X1	(A * B * !D * !Y)	0.00000	0.00000	0.00000
	(A * B * !D * !Y)	15.21310	15.21640	15.42250
	(A * !B * !Y) + (!A * !Y)	0.00000	0.00000	0.00000
	(A * !B * !Y) + (!A * !Y)	1.97856	1.97820	1.97832

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOA4X1	(A * B * !D * !Y)	0.00000	0.00000	0.00000
	(A * B * !D * !Y)	9.30713	9.31369	9.51351
	(A * !B * !Y) + (!A * !Y)	0.00000	0.00000	0.00000
	(A * !B * !Y) + (!A * !Y)	5.00996	5.00981	5.00999

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOA4X1	(A * B * C * !Y) + (A * !B * !Y) + (!A * !Y)	0.00000	0.00000	0.00000
	(A * B * C * !Y) + (A * !B * !Y) + (!A * !Y)	14.46380	14.46260	14.46360

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOA4X1	(A * B * C * !Y) + (A * !B * !Y) + (!A * !Y)	0.00000	0.00000	0.00000
	(A * B * C * !Y) + (A * !B * !Y) + (!A * !Y)	15.10380	15.10340	15.10340

AOAI4X1

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Truth Table

INPUT				OUTPUT
A	**B**	**C**	**D**	**YN**
0	x	x	x	1
1	0	x	x	1
1	1	x	0	1
1	1	0	1	0
1	1	1	1	1

Footprint

Cell Name	Area
AOAI4X1	85.12720

Pin Capacitance Information

Cell Name	Pin Cap(pf)				Max Cap(pf)
	A	**B**	**C**	**D**	**YN**
AOAI4X1	0.01012	0.01034	0.01036	0.01034	4.85021

Leakage Information

Cell Name	Leakage(nW)		
	Min.	**Avg**	**Max.**
AOAI4X1	0.00000	5.05154	26.53960

Delay Information

Delay(ns) to YN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AOAI4X1	A->YN (FR)	0.10666	0.45886	6.75998
	B->YN (FR)	0.10079	0.43774	6.45570
	C->YN (RR)	0.05181	0.34247	5.94709
	D->YN (FR)	0.01893	0.46817	9.73697

Delay(ns) to YN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AOAI4X1	A->YN (RF)	0.15390	0.50606	7.34009
	B->YN (RF)	0.14564	0.49676	7.16256
	C->YN (FF)	0.08696	0.49412	8.28647
	D->YN (RF)	0.02103	0.47128	10.09600

Power Information

Internal switching power(pJ) to YN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
AOAI4X1	A	0.00000	0.00000	0.00000
	A	1.33400	1.34451	1.60016
	B	0.00000	0.00000	0.00000
	B	1.32543	1.33447	1.56399
	C	0.00000	0.00000	0.00000
	C	9.89330	9.89881	10.11460
	D	0.00000	0.00000	0.00000
	D	5.16858	5.17143	5.21974

Internal switching power(pJ) to YN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
AOAI4X1	A	0.00000	0.00000	0.00000
	A	15.43850	15.44250	15.63500
	B	0.00000	0.00000	0.00000
	B	15.44140	15.44590	15.64820
	C	0.00000	0.00000	0.00000
	C	16.30500	16.31480	16.52900
	D	0.00000	0.00000	0.00000
	D	16.33010	16.33250	16.35100

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOAI4X1	(B * C * YN)	0.00000	0.00000	0.00000
	(B * C * YN)	10.88330	10.88830	11.11090
	(B * !C * !D * YN)	0.00000	0.00000	0.00000
	(B * !C * !D * YN)	4.22351	4.22374	4.40153
	(!B * YN)	0.00000	0.00000	0.00000
	(!B * YN)	0.03325	0.03323	0.03320

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOAI4X1	(B * C * YN)	0.00000	0.00000	0.00000
	(B * C * YN)	0.07049	0.08017	0.34734
	(B * !C * !D * YN)	0.00000	0.00000	0.00000
	(B * !C * !D * YN)	1.41009	1.41906	1.66160
	(!B * YN)	0.00000	0.00000	0.00000
	(!B * YN)	0.05468	0.05410	0.05406

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOAI4X1	(A * C * YN)	0.00000	0.00000	0.00000
	(A * C * YN)	10.88370	10.88930	11.11860
	(A * !C * !D * YN)	0.00000	0.00000	0.00000
	(A * !C * !D * YN)	4.22440	4.22579	4.41116
	(!A * YN)	0.00000	0.00000	0.00000
	(!A * YN)	0.03532	0.03527	0.03531

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOAI4X1	(A * C * YN)	0.00000	0.00000	0.00000
	(A * C * YN)	0.06215	0.07043	0.29930
	(A * !C * !D * YN)	0.00000	0.00000	0.00000
	(A * !C * !D * YN)	1.40193	1.40972	1.62390
	(!A * YN)	0.00000	0.00000	0.00000
	(!A * YN)	0.05438	0.05457	0.05452

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOAI4X1	(A * B * !D * YN)	0.00000	0.00000	0.00000
	(A * B * !D * YN)	10.82360	10.82640	11.03450
	(A * !B * YN) + (!A * YN)	0.00000	0.00000	0.00000
	(A * !B * YN) + (!A * YN)	0.02653	0.02651	0.02651

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOAI4X1	(A * B * !D * YN)	0.00000	0.00000	0.00000
	(A * B * !D * YN)	5.42317	5.43017	5.63358
	(A * !B * YN) + (!A * YN)	0.00000	0.00000	0.00000
	(A * !B * YN) + (!A * YN)	1.37252	1.37248	1.37266

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOAI4X1	(A * B * C * YN) + (A * !B * YN) + (!A * YN)	0.00000	0.00000	0.00000
	(A * B * C * YN) + (A * !B * YN) + (!A * YN)	10.13070	10.12750	10.13040

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOAI4X1	(A * B * C * YN) + (A * !B * YN) + (!A * YN)	0.00000	0.00000	0.00000
	(A * B * C * YN) + (A * !B * YN) + (!A * YN)	10.75630	10.75580	10.75600

AOI3X1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT			OUTPUT
A	B	C	YN
0	x	x	0
1	0	x	0
1	1	0	1
1	1	1	0

Footprint

Cell Name	Area
AOI3X1	59.28640

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	A	B	C	YN
AOI3X1	0.01023	0.01045	0.01033	2.77552

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
AOI3X1	0.00000	4.61707	20.74570

Delay Information

Delay(ns) to YN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AOI3X1	A->YN (RR)	0.08848	0.49138	7.21245
	B->YN (RR)	0.08214	0.49669	7.29355
	C->YN (FR)	0.02638	0.53653	9.99639

Delay(ns) to YN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AOI3X1	A->YN (FF)	0.06292	0.31132	3.72465
	B->YN (FF)	0.05847	0.29702	3.47353
	C->YN (RF)	0.01472	0.29856	5.47197

Power Information

Internal switching power(pJ) to YN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
AOI3X1	A	0.00000	0.00000	0.00000
	A	5.57836	5.58359	5.81730
	B	0.00000	0.00000	0.00000
	B	5.58045	5.58659	5.82462
	C	0.00000	0.00000	0.00000
	C	6.78359	6.78732	6.84183

Internal switching power(pJ) to YN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
AOI3X1	A	0.00000	0.00000	0.00000
	A	1.63303	1.64506	1.95919
	B	0.00000	0.00000	0.00000
	B	1.62429	1.63377	1.88611
	C	0.00000	0.00000	0.00000
	C	12.73780	12.74000	12.77120

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOI3X1	(B * C * !YN)	0.00000	0.00000	0.00000
	(B * C * !YN)	13.00610	13.01320	13.26510
	(!B * !YN)	0.00000	0.00000	0.00000
	(!B * !YN)	0.03503	0.03500	0.03499

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOI3X1	(B * C * !YN)	0.00000	0.00000	0.00000
	(B * C * !YN)	0.06326	0.07166	0.31845
	(!B * !YN)	0.00000	0.00000	0.00000
	(!B * !YN)	0.06094	0.06038	0.06031

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOI3X1	(A * C * !YN)	0.00000	0.00000	0.00000
	(A * C * !YN)	13.00770	13.01410	13.26670
	(!A * !YN)	0.00000	0.00000	0.00000
	(!A * !YN)	0.03706	0.03701	0.03698

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOI3X1	(A * C * !YN)	0.00000	0.00000	0.00000
	(A * C * !YN)	0.05485	0.06190	0.26759
	(!A * !YN)	0.00000	0.00000	0.00000
	(!A * !YN)	0.06084	0.06101	0.06098

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOI3X1	(A * !B * !YN) + (!A * !YN)	0.00000	0.00000	0.00000
	(A * !B * !YN) + (!A * !YN)	0.01827	0.01823	0.01823

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOI3X1	**(A * !B * !YN) + (!A * !YN)**	0.00000	0.00000	0.00000
	(A * !B * !YN) + (!A * !YN)	1.60666	1.60656	1.60669

BUFX1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT	OUTPUT
A	**Y**
0	**0**
1	**1**

Footprint

Cell Name	Area
BUFX1	42.05920

Pin Capacitance Information

Cell Name	Pin Cap(pf)	Max Cap(pf)
	A	**Y**
BUFX1	0.01061	5.69256

Leakage Information

Cell Name	Leakage(nW)		
	Min.	**Avg**	**Max.**
BUFX1	0.00000	6.07078	6.07078

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
BUFX1	A->Y (RR)	0.03901	0.33916	6.85692

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
BUFX1	A->Y (FF)	0.04550	0.33988	6.04713

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
BUFX1	A	0.00000	0.00000	0.00000
	A	3.51949	3.53290	3.87490

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
BUFX1	A	0.00000	0.00000	0.00000
	A	3.54116	3.55466	3.91021

DFFQNX1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT		OUTPUT
D	CLK	QN
0	R	1
1	R	0
x	x	IQN

Footprint

Cell Name	Area
DFFQNX1	174.13440

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	D	CLK	QN
DFFQNX1	0.01036	0.02277	4.97752

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFQNX1	0.00000	28.46890	43.32340

Delay Information

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFQNX1	**CLK->QN (RR)**	0.14851	0.49761	6.83008

Delay(ns) to QN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFQNX1	**CLK->QN (RF)**	0.16997	0.57011	7.70865

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFQNX1	hold	CLK (R)	0.04988	0.08001	0.78220
	setup	CLK (R)	0.10873	0.14936	0.75903

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFQNX1	hold	CLK (R)	-0.03460	-0.07672	-0.69255
	setup	CLK (R)	0.06667	0.11431	1.54876

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFQNX1	min_pulse_width	CLK ()	D	0.10887	0.47852	16.50020
	min_pulse_width	CLK ()	!D	0.13786	0.47852	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFQNX1	min_pulse_width	CLK ()	D	0.19585	0.47852	16.50020
	min_pulse_width	CLK ()	!D	0.09800	0.47852	16.50020

Power Information

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFQNX1	CLK	0.00000	0.00000	0.00000
	CLK	26.25550	26.25810	26.59910

Internal switching power(pJ) to QN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFQNX1	CLK	0.00000	0.00000	0.00000
	CLK	21.12960	21.13010	21.34430

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFQNX1	(CLK * QN)	0.00000	0.00000	0.00000
	(CLK * QN)	24.93390	24.93540	24.93730
	(CLK * !QN)	0.00000	0.00000	0.00000
	(CLK * !QN)	21.09450	21.09730	21.31180
	!CLK	0.00000	0.00000	0.00000
	!CLK	11.29550	11.29990	11.51240

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFQNX1	(CLK * QN)	0.00000	0.00000	0.00000
	(CLK * QN)	25.56890	25.56880	25.56880
	(CLK * !QN)	0.00000	0.00000	0.00000
	(CLK * !QN)	10.10200	10.10730	10.33950
	!CLK	0.00000	0.00000	0.00000
	!CLK	11.30940	11.31480	11.53240

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFQNX1	(D * !QN)	0.00000	0.00000	0.00000
	(D * !QN)	21.09760	21.09640	21.30380
	(!D * QN)	0.00000	0.00000	0.00000
	(!D * QN)	26.21410	26.21520	26.41230

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFQNX1	(D * QN)	0.00000	0.00000	0.00000
	(D * QN)	10.32910	10.33160	10.54020
	(D * !QN)	0.00000	0.00000	0.00000
	(D * !QN)	10.11720	10.11860	10.34480
	(!D * QN)	0.00000	0.00000	0.00000
	(!D * QN)	9.90828	9.91134	10.11390
	(!D * !QN)	0.00000	0.00000	0.00000
	(!D * !QN)	11.32300	11.32600	11.54160

DFFQX1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT		OUTPUT
D	CLK	Q
x	x	-

Footprint

Cell Name	Area
DFFQX1	174.13440

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	D	CLK	Q
DFFQX1	0.01038	0.02299	7.52044

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFQX1	0.00000	17.32250	28.05390

Delay Information
Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFQX1	CLK->Q (RR)	0.09059	0.44803	7.64496

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFQX1	hold	CLK (R)	0.04602	0.07009	0.90532
	setup	CLK (R)	0.12154	0.18538	0.99377

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFQX1	hold	CLK (R)	-0.02298	-0.04251	-0.38282
	setup	CLK (R)	9999999999999999635896294965248.00000	9999999999999999635896294965248.00000	9999999999999999635896294965248.00000

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFQX1	min_pulse_width	CLK ()	D	9999999999999999635896294965248.00000	9999999999999999635896294965248.00000	9999999999999999635896294965248.00000

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFQX1	min_pulse_width	CLK ()	D	0.20309	0.47852	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFQX1	CLK	0.00000	0.00000	0.00000
	CLK	62535.30000	62950.40000	63355.10000

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFQX1	(CLK * Q)	0.00000	0.00000	0.00000
	(CLK * Q)	63893.30000	63891.90000	63907.60000
	(CLK * !Q)	0.00000	0.00000	0.00000
	(CLK * !Q)	80310.90000	80311.50000	80311.90000
	(!CLK * !Q)	0.00000	0.00000	0.00000
	(!CLK * !Q)	146380.00000	146380.00000	146387.00000

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFQX1	(CLK * Q)	0.00000	0.00000	0.00000
	(CLK * Q)	65149.30000	65148.80000	65150.10000
	(CLK * !Q)	0.00000	0.00000	0.00000
	(CLK * !Q)	79220.00000	79220.60000	79220.70000
	(!CLK * !Q)	0.00000	0.00000	0.00000
	(!CLK * !Q)	145918.00000	145922.00000	145934.00000

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFQX1	(!D * !Q)	0.00000	0.00000	0.00000
	(!D * !Q)	80089.70000	80088.60000	80092.00000

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFQX1	(D * !Q)	0.00000	0.00000	0.00000
	(D * !Q)	157385.00000	157380.00000	157372.00000
	(!D * !Q)	0.00000	0.00000	0.00000
	(!D * !Q)	157183.00000	157177.00000	157178.00000

DFFRNQNX1

sky130_rhbd_tt_1P8_25C.ccs Cell Library:
Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT			OUTPUT
D	RN	CLK	QN
0	1	R	1
1	1	R	0
x	0	x	1
x	1	x	IQN

Footprint

Cell Name	Area
DFFRNQNX1	208.58881

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	D	RN	CLK	QN
DFFRNQNX1	0.01008	0.03153	0.02307	7.15843

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFRNQNX1	0.00000	29.67510	40.69830

Delay Information

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFRNQNX1	CLK->QN (RR)	0.13499	0.47937	8.08564
	RN->QN (FR)	0.02749	0.45618	10.75420

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFRNQNX1	hold	CLK (R)	0.04021	0.05233	1.00038
	setup	CLK (R)	99999999999999996358962949652480.00000	99999999999999996358962949652480.00000	99999999999999996358962949652480.00000

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFRNQNX1	hold	CLK (R)	-0.00568	-0.04251	-0.41174
	setup	CLK (R)	0.07540	0.12167	1.80218

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQNX1	hold	CLK (R)	RN	0.04021	0.05233	1.00038
	setup	CLK (R)	RN	99999999999999996358962949652480.00000	99999999999999996358962949652480.00000	99999999999999996358962949652480.00000

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQNX1	hold	CLK (R)	RN	-0.00568	-0.04251	-0.41174
	setup	CLK (R)	RN	0.07540	0.12167	1.80218

Constraints(ns) for RN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQNX1	recovery	CLK (R)	D	0.15446	0.22928	7.76285
	removal	CLK (R)	D	0.03631	0.04251	0.01898

Constraints(ns) for RN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQNX1	min_pulse_width	RN ()	(CLK * D)	0.12337	0.47852	16.50020
	min_pulse_width	RN ()	(CLK * !D)	0.09075	0.47852	16.50020
	min_pulse_width	RN ()	(!CLK * D)	99999999999999996358962949652480.00000	99999999999999996358962949652480.00000	99999999999999996358962949652480.00000
	min_pulse_width	RN ()	(!CLK * !D)	99999999999999996358962949652480.00000	99999999999999996358962949652480.00000	99999999999999996358962949652480.00000

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQNX1	min_pulse_width	CLK ()	(!D * RN)	99999999999999996358962949652480.00000	99999999999999996358962949652480.00000	99999999999999996358962949652480.00000

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQNX1	min_pulse_width	CLK ()	(!D * RN)	0.10525	0.47852	16.50020

Power Information

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFRNQNX1	CLK	0.00000	0.00000	0.00000
	CLK	24.93040	24.95650	25.67930
	RN	25.04020	25.06400	25.59500

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQNX1	(CLK * RN * QN)	0.00000	0.00000	0.00000
	(CLK * RN * QN)	22.76230	22.76270	22.76390
	(CLK * RN * !QN)	0.00000	0.00000	0.00000
	(CLK * RN * !QN)	57746.60000	57746.00000	57747.20000
	(CLK * !RN * QN)	0.00000	0.00000	0.00000
	(CLK * !RN * QN)	22.74390	22.74700	22.74890
	(!CLK * RN * !QN)	0.00000	0.00000	0.00000
	(!CLK * RN * !QN)	80724.30000	80723.20000	80726.90000
	(!CLK * !RN * QN)	0.00000	0.00000	0.00000
	(!CLK * !RN * QN)	9.86636	9.86613	9.86654

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQNX1	(CLK * RN * QN)	0.00000	0.00000	0.00000
	(CLK * RN * QN)	23.37400	23.37500	23.37510
	(CLK * RN * !QN)	0.00000	0.00000	0.00000
	(CLK * RN * !QN)	60607.00000	60607.90000	60608.10000
	(CLK * !RN * QN)	0.00000	0.00000	0.00000
	(CLK * !RN * QN)	23.36600	23.36660	23.36690
	(!CLK * RN * !QN)	0.00000	0.00000	0.00000
	(!CLK * RN * !QN)	79550.60000	79552.50000	79551.00000
	(!CLK * !RN * QN)	0.00000	0.00000	0.00000
	(!CLK * !RN * QN)	10.43450	10.43420	10.43400

Passive power(pJ) for RN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQNX1	(CLK * D * QN)	0.00000	0.00000	0.00000
	(CLK * D * QN)	22.01890	22.01690	22.01690
	(CLK * !D * QN)	0.00000	0.00000	0.00000
	(CLK * !D * QN)	22.01070	22.00840	22.00990

Passive power(pJ) for RN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQNX1	(CLK * D * QN)	0.00000	0.00000	0.00000
	(CLK * D * QN)	23.85710	23.85770	23.85810
	(CLK * !D * QN)	0.00000	0.00000	0.00000
	(CLK * !D * QN)	23.85700	23.85730	23.85810

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQNX1	(D * RN * !QN)	0.00000	0.00000	0.00000
	(D * RN * !QN)	54869.40000	55190.60000	55677.10000
	(D * !RN * QN)	0.00000	0.00000	0.00000
	(D * !RN * QN)	23.89950	23.89960	24.08480
	(!D * !RN * QN)	0.00000	0.00000	0.00000
	(!D * !RN * QN)	23.89670	23.89710	24.08390

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQNX1	(D * RN * !QN)	0.00000	0.00000	0.00000
	(D * RN * !QN)	76494.90000	76499.40000	76501.80000
	(D * !RN * QN)	0.00000	0.00000	0.00000
	(D * !RN * QN)	9.22269	9.22454	9.41734
	(!D * RN * !QN)	0.00000	0.00000	0.00000
	(!D * RN * !QN)	75304.50000	75312.70000	75328.40000
	(!D * !RN * QN)	0.00000	0.00000	0.00000
	(!D * !RN * QN)	9.22367	9.22719	9.42368

DFFRNQX1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT			OUTPUT
D	RN	CLK	Q
x	x	x	-

Footprint

Cell Name	Area
DFFRNQX1	208.58881

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	D	RN	CLK	Q
DFFRNQX1	0.01000	0.02835	0.02327	7.56534

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFRNQX1	0.00000	27.27400	56.43520

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFRNQX1	**CLK->Q (RR)**	0.14699	0.50276	7.83177

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFRNQX1	hold	CLK (R)	0.03433	0.05029	1.11071
	setup	CLK (R)	0.15339	0.20353	0.80836

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFRNQX1	hold	CLK (R)	0.00070	-0.03501	-0.37993
	setup	CLK (R)	9999999999999999635896294965248.00000	9999999999999999635896294965248.00000	9999999999999999635896294965248.00000

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQX1	hold	CLK (R)	RN	0.03433	0.05029	1.11071
	setup	CLK (R)	RN	0.15339	0.20353	0.80836

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQX1	hold	CLK (R)	RN	0.00070	-0.03501	-0.37993
	setup	CLK (R)	RN	9999999999999999635896294965248.00000	9999999999999999635896294965248.00000	9999999999999999635896294965248.00000

Constraints(ns) for RN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFRNQX1	recovery	CLK (R)	-0.02012	-0.02542	0.27630
	removal	CLK (R)	0.03634	0.04001	-0.00846

Constraints(ns) for RN falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFRNQX1	recovery	CLK (F)	-0.16885	-0.33755	-1.47980
	removal	CLK (F)	-0.10809	-0.24754	0.62429

Constraints(ns) for RN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQX1	removal	CLK (R)	D	0.03634	0.04001	-0.00846
	setup	CLK (R)	D	-0.02012	-0.02542	0.27630

Constraints(ns) for RN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQX1	recovery	CLK (F)	D	-0.16885	-0.33755	-1.47980
	removal	CLK (F)	D	-0.10809	-0.24754	-0.56234
	removal	CLK (F)	!D	-0.11033	-0.26754	0.62429
	setup	CLK (R)	D	-0.04253	-0.14502	-1.29700

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQX1	min_pulse_width	CLK ()	(D * RN)	9999999999999999635896294965248.00000	9999999999999999635896294965248.00000	9999999999999999635896294965248.00000

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQX1	min_pulse_width	CLK ()	(D * RN)	0.23571	0.47852	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFRNQX1	CLK	0.00000	0.00000	0.00000
	CLK	35.07020	35.09750	35.76120

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQX1	(CLK * RN * Q)	0.00000	0.00000	0.00000
	(CLK * RN * Q)	34.33070	34.33170	34.52260
	(CLK * RN * !Q)	0.00000	0.00000	0.00000
	(CLK * RN * !Q)	80688.50000	80690.20000	80689.50000
	(CLK * !RN * !Q)	0.00000	0.00000	0.00000
	(CLK * !RN * !Q)	79124.70000	79125.60000	79125.00000
	(!CLK * RN * !Q)	0.00000	0.00000	0.00000
	(!CLK * RN * !Q)	83869.90000	83868.90000	83866.90000
	(!CLK * !RN * !Q)	0.00000	0.00000	0.00000
	(!CLK * !RN * !Q)	78478.50000	78477.80000	78479.40000

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQX1	(CLK * RN * Q)	0.00000	0.00000	0.00000
	(CLK * RN * Q)	16.54240	16.54820	16.76080
	(CLK * RN * !Q)	0.00000	0.00000	0.00000
	(CLK * RN * !Q)	80034.40000	80035.80000	80035.30000
	(CLK * !RN * !Q)	0.00000	0.00000	0.00000
	(CLK * !RN * !Q)	77903.50000	77904.30000	77903.50000
	(!CLK * RN * !Q)	0.00000	0.00000	0.00000
	(!CLK * RN * !Q)	84083.30000	84081.70000	84083.30000
	(!CLK * !RN * !Q)	0.00000	0.00000	0.00000
	(!CLK * !RN * !Q)	78751.10000	78764.20000	78776.50000

Passive power(pJ) for RN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQX1	(CLK * D * !Q)	0.00000	0.00000	0.00000
	(CLK * D * !Q)	80706.10000	80708.70000	80707.60000
	(CLK * !D * !Q)	0.00000	0.00000	0.00000
	(CLK * !D * !Q)	80400.70000	80402.50000	80401.90000
	(!CLK * D * !Q)	0.00000	0.00000	0.00000
	(!CLK * D * !Q)	84184.30000	84183.60000	84183.30000
	(!CLK * !D * !Q)	0.00000	0.00000	0.00000
	(!CLK * !D * !Q)	84156.70000	84155.40000	84156.20000

Passive power(pJ) for RN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQX1	(CLK * D * !Q)	0.00000	0.00000	0.00000
	(CLK * D * !Q)	77589.90000	77592.90000	77596.40000
	(CLK * !D * !Q)	0.00000	0.00000	0.00000
	(CLK * !D * !Q)	77225.80000	77227.70000	77232.90000
	(!CLK * D * !Q)	0.00000	0.00000	0.00000
	(!CLK * D * !Q)	77745.20000	77733.80000	77754.60000
	(!CLK * !D * !Q)	0.00000	0.00000	0.00000
	(!CLK * !D * !Q)	77971.90000	77973.00000	77972.70000

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQX1	(D * !RN * !Q)	0.00000	0.00000	0.00000
	(D * !RN * !Q)	78168.40000	78179.90000	78220.80000
	(!D * RN * !Q)	0.00000	0.00000	0.00000
	(!D * RN * !Q)	80197.20000	80198.50000	80201.10000
	(!D * !RN * !Q)	0.00000	0.00000	0.00000
	(!D * !RN * !Q)	78244.80000	78257.20000	78312.40000

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQX1	(D * RN * !Q)	0.00000	0.00000	0.00000
	(D * RN * !Q)	83151.20000	83082.50000	82951.00000
	(D * !RN * !Q)	0.00000	0.00000	0.00000
	(D * !RN * !Q)	78191.80000	78191.60000	78191.60000
	(!D * RN * !Q)	0.00000	0.00000	0.00000
	(!D * RN * !Q)	83952.70000	83952.40000	83951.80000
	(!D * !RN * !Q)	0.00000	0.00000	0.00000
	(!D * !RN * !Q)	78718.80000	78718.90000	78717.60000

DFFRNX1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT			OUTPUT	
D	RN	CLK	Q	QN
0	1	R	0	1
1	1	R	1	0
x	0	x	0	1
x	1	x	IQ	IQN

Footprint

Cell Name	Area
DFFRNX1	208.58881

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)	
	D	RN	CLK	Q	QN
DFFRNX1	0.00999	0.03117	0.02307	5.03685	3.49563

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFRNX1	0.00000	33.26000	56.43530

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFRNX1	CLK->Q (RR)	0.16902	0.54557	7.02121
	QN->Q (FR)	0.03896	0.49909	10.12640

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFRNX1	CLK->Q (RF)	0.21923	0.95341	17.17050
	QN->Q (RF)	0.04434	0.46142	9.77994
	RN->Q (FF)	0.09376	0.95370	20.72680

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFRNX1	CLK->QN (RR)	0.16266	0.46896	5.12849
	Q->QN (FR)	0.03536	0.44523	8.21535
	RN->QN (FR)	0.03785	0.45743	8.34531

Delay(ns) to QN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFRNX1	CLK->QN (RF)	0.23817	1.00220	15.20000
	Q->QN (RF)	0.05233	0.53046	9.99979

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFRNX1	hold	CLK (R)	0.03824	0.05385	0.78265
	setup	CLK (R)	0.14672	0.18812	0.64770

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFRNX1	hold	CLK (R)	-0.02439	-0.06601	-0.66455
	setup	CLK (R)	0.06787	0.11580	1.79623

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNX1	hold	CLK (R)	RN	0.03824	0.05385	0.78265
	setup	CLK (R)	RN	0.14672	0.18812	0.64770

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNX1	hold	CLK (R)	RN	-0.02439	-0.06601	-0.66455
	setup	CLK (R)	RN	0.06787	0.11580	1.79623

Constraints(ns) for RN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFRNX1	recovery	CLK (R)	0.13818	0.19844	7.21987
	removal	CLK (R)	0.03631	0.04251	0.02043

Constraints(ns) for RN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNX1	recovery	CLK (R)	D	0.13818	0.19844	7.21987
	removal	CLK (R)	D	0.03631	0.04251	0.02043

Constraints(ns) for RN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNX1	min_pulse_width	RN ()	(CLK * D)	0.12337	0.47852	16.50020
	min_pulse_width	RN ()	(CLK * !D)	0.09075	0.47852	16.50020
	min_pulse_width	RN ()	(!CLK * D)	0.05814	0.47852	16.50020
	min_pulse_width	RN ()	(!CLK * !D)	0.05814	0.47852	16.50020

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNX1	min_pulse_width	CLK ()	(D * RN)	0.16686	0.47852	16.50020
	min_pulse_width	CLK ()	(!D * RN)	0.14874	0.47852	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNX1	min_pulse_width	CLK ()	(D * RN)	0.23571	0.47852	16.50020
	min_pulse_width	CLK ()	(!D * RN)	0.10525	0.47852	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFRNX1	CLK	0.00000	0.00000	0.00000
	CLK	17.25750	17.25740	17.34950

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFRNX1	CLK	0.00000	0.00000	0.00000
	CLK	11.89100	11.89280	11.99550
	RN	-0.00114	-0.15445	-4.07983
	RN	12.11450	12.11880	12.22040

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFRNX1	CLK	0.00000	0.00000	0.00000
	CLK	11.89130	11.89300	11.99420
	RN	-0.00114	-0.12405	-2.83145
	RN	12.11450	12.11900	12.22200

Internal switching power(pJ) to QN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFRNX1	CLK	0.00000	0.00000	0.00000
	CLK	17.25740	17.25780	17.35280

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNX1	(CLK * RN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * RN * Q * !QN)	34.32470	34.32660	34.51870
	(CLK * RN * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * RN * !Q * QN)	22.75670	22.75610	22.75740
	(CLK * !RN * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * !RN * !Q * QN)	22.74060	22.74200	22.74390
	(!CLK * RN * Q * !QN) + (!CLK * RN * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * RN * Q * !QN) + (!CLK * RN * !Q * QN)	18.84230	18.84350	19.03410
	(!CLK * !RN * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * !RN * !Q * QN)	9.85929	9.85841	9.85956

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNX1	(CLK * RN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * RN * Q * !QN)	16.53670	16.54180	16.75820
	(CLK * RN * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * RN * !Q * QN)	23.36850	23.36860	23.36900
	(CLK * !RN * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * !RN * !Q * QN)	23.36100	23.36130	23.36160
	(!CLK * RN * Q * !QN) + (!CLK * RN * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * RN * Q * !QN) + (!CLK * RN * !Q * QN)	13.88310	13.88820	14.10180
	(!CLK * !RN * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * !RN * !Q * QN)	10.43010	10.42930	10.42920

Passive power(pJ) for RN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNX1	(CLK * D * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * D * !Q * QN)	22.01640	22.01270	22.01410
	(CLK * !D * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * !D * !Q * QN)	22.00690	22.00530	22.00640
	(!CLK * D * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * D * !Q * QN)	13.81720	13.82080	14.01690
	(!CLK * !D * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * !D * !Q * QN)	9.22439	9.22474	9.22417

Passive power(pJ) for RN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNX1	(CLK * D * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * D * !Q * QN)	23.85130	23.85190	23.85260
	(CLK * !D * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * !D * !Q * QN)	23.84730	23.85260	23.85340
	(!CLK * D * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * D * !Q * QN)	10.54300	10.54650	10.74120
	(!CLK * !D * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * !D * !Q * QN)	10.91300	10.91190	10.91150

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNX1	(D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(D * RN * Q * !QN)	34.32710	34.32340	34.50840
	(D * !RN * !Q * QN)	0.00000	0.00000	0.00000
	(D * !RN * !Q * QN)	23.89340	23.89370	24.07910
	(!D * RN * !Q * QN)	0.00000	0.00000	0.00000
	(!D * RN * !Q * QN)	23.90580	23.90550	24.09530
	(!D * !RN * !Q * QN)	0.00000	0.00000	0.00000
	(!D * !RN * !Q * QN)	23.89080	23.89200	24.07890

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNX1	(D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(D * RN * Q * !QN)	16.55210	16.55180	16.76340
	(D * RN * !Q * QN)	0.00000	0.00000	0.00000
	(D * RN * !Q * QN)	13.80650	13.80600	13.99430
	(D * !RN * !Q * QN)	0.00000	0.00000	0.00000
	(D * !RN * !Q * QN)	9.21592	9.21678	9.40970
	(!D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(!D * RN * Q * !QN)	13.90220	13.90700	14.12620
	(!D * RN * !Q * QN)	0.00000	0.00000	0.00000
	(!D * RN * !Q * QN)	9.22134	9.22235	9.42213
	(!D * !RN * !Q * QN)	0.00000	0.00000	0.00000
	(!D * !RN * !Q * QN)	9.21623	9.21887	9.41555

DFFSNQNX1

sky130_rhbd_tt_1P8_25C.ccs Cell Library:
Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT			OUTPUT
D	SN	CLK	QN
x	x	x	-

Footprint

Cell Name	Area
DFFSNQNX1	197.10400

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	D	SN	CLK	QN
DFFSNQNX1	0.01014	0.01981	0.02220	7.76387

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFSNQNX1	0.00000	24.27620	57.62640

Delay Information

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	**Mid**	**Last**
DFFSNQNX1	**CLK->QN (RR)**	0.12284	0.46867	8.38472

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNQNX1	hold	CLK (R)	0.00977	0.01456	1.25830
	setup	CLK (R)	9999999999999999635896294965248.00000	9999999999999999635896294965248.00000	9999999999999999635896294965248.00000

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNQNX1	hold	CLK (R)	-0.05934	-0.10501	-0.92019
	setup	CLK (R)	0.11206	0.17157	1.50404

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQNX1	hold	CLK (R)	SN	0.00977	0.01456	1.25830
	setup	CLK (R)	SN	9999999999999999635896294965248.00000	9999999999999999635896294965248.00000	9999999999999999635896294965248.00000

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQNX1	hold	CLK (R)	SN	-0.05934	-0.10501	-0.92019
	setup	CLK (R)	SN	0.11206	0.17157	1.50404

Constraints(ns) for SN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNQNX1	removal	CLK (R)	0.00338	0.03000	0.04591

Constraints(ns) for SN falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNQNX1	removal	CLK (F)	-0.08219	-0.21753	-0.39030

Constraints(ns) for SN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQNX1	recovery	CLK (R)	!D	0.02846	0.03110	1.41080
	removal	CLK (R)	!D	0.00338	0.03000	0.04591

Constraints(ns) for SN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQNX1	recovery	CLK (F)	!D	-0.07558	-0.19753	-0.55272
	removal	CLK (F)	D	-0.08219	-0.21753	-0.39523
	removal	CLK (F)	!D	-0.08219	-0.21753	-0.39030

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQNX1	min_pulse_width	CLK ()	(!D * SN)	9999999999999999635896294965248.00000	9999999999999999635896294965248.00000	9999999999999999635896294965248.00000

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQNX1	min_pulse_width	CLK ()	(!D * SN)	0.11249	0.47852	16.50020

Power Information

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNQNX1	CLK	0.00000	0.00000	0.00000
	CLK	35.92480	35.95130	36.69020

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQNX1	(CLK * SN * QN)	0.00000	0.00000	0.00000
	(CLK * SN * QN)	33.71860	33.72010	33.72070
	(CLK * SN * !QN) + (!CLK * !SN * !QN)	0.00000	0.00000	0.00000
	(CLK * SN * !QN) + (!CLK * !SN * !QN)	64688.80000	64689.90000	64704.70000
	(CLK * !SN * !QN)	0.00000	0.00000	0.00000
	(CLK * !SN * !QN)	66067.70000	66089.00000	66145.40000
	(!CLK * SN * !QN)	0.00000	0.00000	0.00000
	(!CLK * SN * !QN)	77620.40000	77615.80000	77615.80000

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQNX1	(CLK * SN * QN)	0.00000	0.00000	0.00000
	(CLK * SN * QN)	34.36370	34.36480	34.36470
	(CLK * SN * !QN) + (!CLK * !SN * !QN)	0.00000	0.00000	0.00000
	(CLK * SN * !QN) + (!CLK * !SN * !QN)	67465.40000	67467.70000	67468.30000
	(CLK * !SN * !QN)	0.00000	0.00000	0.00000
	(CLK * !SN * !QN)	68499.20000	68499.80000	68499.30000
	(!CLK * SN * !QN)	0.00000	0.00000	0.00000
	(!CLK * SN * !QN)	78285.20000	78285.00000	78281.00000

Passive power(pJ) for SN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQNX1	(CLK * D * !QN)	0.00000	0.00000	0.00000
	(CLK * D * !QN)	69023.20000	69024.10000	69026.10000
	(CLK * !D * !QN)	0.00000	0.00000	0.00000
	(CLK * !D * !QN)	68218.10000	68219.90000	68220.20000
	(!CLK * D * !QN)	0.00000	0.00000	0.00000
	(!CLK * D * !QN)	75578.50000	75574.60000	75577.70000
	(!CLK * !D * !QN)	0.00000	0.00000	0.00000
	(!CLK * !D * !QN)	76264.60000	76258.90000	76263.30000

Passive power(pJ) for SN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQNX1	(CLK * D * !QN)	0.00000	0.00000	0.00000
	(CLK * D * !QN)	65918.20000	65919.40000	65918.20000
	(CLK * !D * !QN)	0.00000	0.00000	0.00000
	(CLK * !D * !QN)	67056.70000	67065.20000	67067.80000
	(!CLK * D * !QN)	0.00000	0.00000	0.00000
	(!CLK * D * !QN)	68264.00000	68265.70000	68266.10000
	(!CLK * !D * !QN)	0.00000	0.00000	0.00000
	(!CLK * !D * !QN)	69163.60000	69164.20000	69164.90000

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQNX1	(D * SN * !QN)	0.00000	0.00000	0.00000
	(D * SN * !QN)	64515.20000	64518.50000	64537.20000
	(D * !SN * !QN)	0.00000	0.00000	0.00000
	(D * !SN * !QN)	66548.50000	66550.80000	66562.50000
	(!D * !SN * !QN)	0.00000	0.00000	0.00000
	(!D * !SN * !QN)	64606.00000	64605.30000	64620.90000

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQNX1	(D * SN * !QN)	0.00000	0.00000	0.00000
	(D * SN * !QN)	74455.90000	74455.10000	74453.70000
	(D * !SN * !QN)	0.00000	0.00000	0.00000
	(D * !SN * !QN)	71336.60000	71338.00000	71339.00000
	(!D * SN * !QN)	0.00000	0.00000	0.00000
	(!D * SN * !QN)	74883.40000	74879.60000	74888.00000
	(!D * !SN * !QN)	0.00000	0.00000	0.00000
	(!D * !SN * !QN)	71254.10000	71254.80000	71255.20000

DFFSNQX1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT			OUTPUT
D	SN	CLK	Q
0	1	R	0
1	1	R	1
x	0	x	1
x	1	x	IQ

Footprint

Cell Name	Area
DFFSNQX1	197.10400

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	D	SN	CLK	Q
DFFSNQX1	0.01000	0.02135	0.02197	3.45759

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFSNQX1	0.00000	30.10450	57.62650

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNQX1	CLK->Q (RR)	0.11693	0.43340	4.89334
	SN->Q (FR)	0.03860	0.46137	8.38974

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNQX1	CLK->Q (RF)	0.22620	0.64269	7.66046

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNQX1	hold	CLK (R)	0.01292	0.01544	0.79023
	setup	CLK (R)	0.14021	0.19245	3.23897

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNQX1	hold	CLK (R)	-0.07203	-0.13252	-1.11442
	setup	CLK (R)	0.10429	0.16918	1.57983

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQX1	hold	CLK (R)	SN	0.01292	0.01544	0.79023
	setup	CLK (R)	SN	0.14021	0.19245	3.23897

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQX1	hold	CLK (R)	SN	-0.07203	-0.13252	-1.11442
	setup	CLK (R)	SN	0.10429	0.16918	1.57983

Constraints(ns) for SN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNQX1	recovery	CLK (R)	0.03458	0.03601	4.26577
	removal	CLK (R)	-0.01435	-0.01500	-0.11230

Constraints(ns) for SN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQX1	recovery	CLK (R)	!D	0.03458	0.03601	4.26577
	removal	CLK (R)	!D	-0.01435	-0.01500	-0.11230

Constraints(ns) for SN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQX1	min_pulse_width	SN ()	(CLK * D)	0.07263	0.47852	16.50020
	min_pulse_width	SN ()	(CLK * !D)	0.06901	0.47852	16.50020
	min_pulse_width	SN ()	(!CLK * D)	0.06538	0.47852	16.50020
	min_pulse_width	SN ()	(!CLK * !D)	0.06538	0.47852	16.50020

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQX1	min_pulse_width	CLK ()	(D * SN)	0.10887	0.47852	16.50020
	min_pulse_width	CLK ()	(!D * SN)	0.13786	0.47852	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQX1	min_pulse_width	CLK ()	(D * SN)	0.21759	0.47852	16.50020
	min_pulse_width	CLK ()	(!D * SN)	0.10887	0.47852	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNQX1	CLK	0.00000	0.00000	0.00000
	CLK	19.63960	19.64420	19.95180
	SN	18.99580	19.00300	19.26000

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNQX1	CLK	0.00000	0.00000	0.00000
	CLK	35.36140	35.36530	35.57110

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQX1	(CLK * SN * Q) + (!CLK * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * SN * Q) + (!CLK * !SN * Q)	19.80280	19.80400	20.00250
	(CLK * SN * !Q)	0.00000	0.00000	0.00000
	(CLK * SN * !Q)	33.71200	33.71320	33.71330
	(CLK * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * !SN * Q)	19.79880	19.80020	20.00030
	(!CLK * SN)	0.00000	0.00000	0.00000
	(!CLK * SN)	14.27320	14.27680	14.47470

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQX1	(CLK * SN * Q) + (!CLK * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * SN * Q) + (!CLK * !SN * Q)	9.55580	9.55970	9.76763
	(CLK * SN * !Q)	0.00000	0.00000	0.00000
	(CLK * SN * !Q)	34.35860	34.35890	34.35880
	(CLK * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * !SN * Q)	9.55872	9.56229	9.77805
	(!CLK * SN)	0.00000	0.00000	0.00000
	(!CLK * SN)	19.42250	19.42900	19.63780

Passive power(pJ) for SN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQX1	(CLK * D * Q)	0.00000	0.00000	0.00000
	(CLK * D * Q)	18.32360	18.33020	18.33160
	(CLK * !D * Q)	0.00000	0.00000	0.00000
	(CLK * !D * Q)	9.86576	9.87165	9.87140
	(!CLK * D * Q)	0.00000	0.00000	0.00000
	(!CLK * D * Q)	9.88029	9.87597	9.88002
	(!CLK * !D * Q)	0.00000	0.00000	0.00000
	(!CLK * !D * Q)	14.40890	14.41020	14.60110

Passive power(pJ) for SN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQX1	(CLK * D * Q)	0.00000	0.00000	0.00000
	(CLK * D * Q)	19.55030	19.55690	19.55800
	(CLK * !D * Q)	0.00000	0.00000	0.00000
	(CLK * !D * Q)	11.02360	11.02650	11.02700
	(!CLK * D * Q)	0.00000	0.00000	0.00000
	(!CLK * D * Q)	11.02770	11.02750	11.02750
	(!CLK * !D * Q)	0.00000	0.00000	0.00000
	(!CLK * !D * Q)	3.67485	3.68074	3.88728

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQX1	(D * SN * Q)	0.00000	0.00000	0.00000
	(D * SN * Q)	19.79590	19.79870	20.01610
	(D * !SN * Q)	0.00000	0.00000	0.00000
	(D * !SN * Q)	19.79270	19.79400	20.00870
	(!D * SN * !Q)	0.00000	0.00000	0.00000
	(!D * SN * !Q)	35.14020	35.13980	35.33860
	(!D * !SN * Q)	0.00000	0.00000	0.00000
	(!D * !SN * Q)	10.63750	10.64750	10.90800

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQX1	(D * SN * Q)	0.00000	0.00000	0.00000
	(D * SN * Q)	9.56798	9.57888	9.84257
	(D * SN * !Q)	0.00000	0.00000	0.00000
	(D * SN * !Q)	13.00230	13.00030	13.24090
	(D * !SN * Q)	0.00000	0.00000	0.00000
	(D * !SN * Q)	9.55914	9.57114	9.81192
	(!D * SN * Q)	0.00000	0.00000	0.00000
	(!D * SN * Q)	14.92540	14.93160	15.17560
	(!D * SN * !Q)	0.00000	0.00000	0.00000
	(!D * SN * !Q)	17.39180	17.39250	17.60400
	(!D * !SN * Q)	0.00000	0.00000	0.00000
	(!D * !SN * Q)	3.48914	3.49604	3.79523

DFFSNRNQNX1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT				OUTPUT
D	RN	SN	CLK	QN
0	1	1	R	1
1	1	1	R	0
x	0	x	x	1
x	1	0	x	0
x	1	1	x	IQN

Footprint

Cell Name	Area
DFFSNRNQNX1	231.55840

Pin Capacitance Information

Cell Name	Pin Cap(pf)				Max Cap(pf)
	D	RN	SN	CLK	QN
DFFSNRNQNX1	0.01000	0.03392	0.02220	0.02197	3.37843

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFSNRNQNX1	0.00000	32.92020	54.41010

Delay Information

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNRNQNX1	CLK->QN (RR)	0.16003	0.46092	4.98198
	RN->QN (FR)	0.03842	0.45636	8.30827

Delay(ns) to QN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNRNQNX1	CLK->QN (RF)	0.23146	0.65002	7.52882
	RN->QN (RF)	0.26092	0.71520	10.21090
	SN->QN (FF)	0.33520	0.76941	8.25145

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNRNQNX1	hold	CLK (R)	-0.01228	-0.00979	2.09282
	setup	CLK (R)	0.16731	0.19838	0.66863

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNRNQNX1	hold	CLK (R)	-0.06904	-0.12402	-1.05234
	setup	CLK (R)	0.11676	0.18136	2.38380

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQNX1	hold	CLK (R)	(RN * SN)	-0.01228	-0.00979	2.09282
	setup	CLK (R)	(RN * SN)	0.16731	0.19838	0.66863

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQNX1	hold	CLK (R)	(RN * SN)	-0.06904	-0.12402	-1.05234
	setup	CLK (R)	(RN * SN)	0.11676	0.18136	2.38380

Constraints(ns) for RN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNRNQNX1	recovery	CLK (R)	0.16016	0.20991	7.38019
	removal	CLK (R)	-0.01427	-0.02250	-0.06490
	hold	SN (R)	0.02137	0.05251	0.42234
	setup	SN (R)	0.02344	0.03751	1.09863

Constraints(ns) for RN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQNX1	recovery	CLK (R)	(D * SN)	0.16016	0.20991	7.38019
	removal	CLK (R)	(D * SN)	-0.01427	-0.02250	-0.06490
	hold	SN (R)	(CLK * D)	0.00387	-0.01250	-0.10723
	hold	SN (R)	(CLK * !D)	0.00387	-0.01250	-0.11378
	hold	SN (R)	(!CLK * D)	0.02137	0.05251	0.41659
	hold	SN (R)	(!CLK * !D)	0.01987	0.05001	0.42234
	setup	SN (R)	(CLK * D)	0.00595	0.02400	0.53973
	setup	SN (R)	(CLK * !D)	0.00877	0.02400	0.64724
	setup	SN (R)	(!CLK * D)	0.02319	0.03751	1.09863
	setup	SN (R)	(!CLK * !D)	0.02344	0.03751	0.91800

Constraints(ns) for RN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQNX1	min_pulse_width	RN ()	(CLK * D * SN)	0.11249	0.47852	16.50020
	min_pulse_width	RN ()	(CLK * !D * SN)	0.10162	0.47852	16.50020
	min_pulse_width	RN ()	(!CLK * D * SN)	0.06538	0.47852	16.50020
	min_pulse_width	RN ()	(!CLK * !D * SN)	0.06538	0.47852	16.50020

Constraints(ns) for SN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNRNQNX1	recovery	CLK (R)	0.03050	0.02462	0.25888
	removal	CLK (R)	0.00269	0.00750	0.01240
	hold	RN (R)	0.04180	0.08001	0.39008
	setup	RN (R)	0.01187	-0.00535	0.64057

Constraints(ns) for SN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQNX1	recovery	CLK (R)	(!D * RN)	0.03050	0.02462	0.25888
	removal	CLK (R)	(!D * RN)	0.00269	0.00750	0.01240
	hold	RN (R)	CLK	0.03494	0.08001	0.39008
	hold	RN (R)	(!CLK * D)	0.04180	0.07751	0.35597
	hold	RN (R)	(!CLK * !D)	0.04180	0.07751	0.36994
	setup	RN (R)	CLK	0.01187	-0.00535	0.64057
	setup	RN (R)	(!CLK * D)	-0.01595	-0.03000	0.39737
	setup	RN (R)	(!CLK * !D)	-0.01818	-0.03000	0.03997

Constraints(ns) for SN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQNX1	min_pulse_width	SN ()	(CLK * D * RN)	0.09075	0.47852	16.50020
	min_pulse_width	SN ()	(CLK * !D * RN)	0.09075	0.47852	16.50020
	min_pulse_width	SN ()	(!CLK * D * RN)	0.07626	0.47852	16.50020
	min_pulse_width	SN ()	(!CLK * !D * RN)	0.07263	0.47852	16.50020

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQNX1	min_pulse_width	CLK ()	(D * RN * SN)	0.15961	0.47852	16.50020
	min_pulse_width	CLK ()	(!D * RN * SN)	0.14874	0.47852	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQNX1	min_pulse_width	CLK ()	(D * RN * SN)	0.23571	0.47852	16.50020
	min_pulse_width	CLK ()	(!D * RN * SN)	0.12699	0.47852	16.50020

Power Information

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNRNQNX1	CLK	0.00000	0.00000	0.00000
	CLK	32.50460	32.50450	32.78980
	RN	33.05860	33.06610	33.34040

Internal switching power(pJ) to QN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNRNQNX1	CLK	0.00000	0.00000	0.00000
	CLK	32.67070	32.67240	32.86050
	RN	32.88900	32.89120	33.16160
	SN	-0.00228	-0.24308	-5.47302
	SN	32.59500	32.60570	32.93920

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQNX1	(CLK * RN * SN * QN)	0.00000	0.00000	0.00000
	(CLK * RN * SN * QN)	31.10850	31.10890	31.10950
	(CLK * RN * SN * !QN) + (!CLK * RN * !SN * !QN)	0.00000	0.00000	0.00000
	(CLK * RN * SN * !QN) + (!CLK * RN * !SN * !QN)	32.63700	32.63790	32.83090
	(CLK * RN * !SN * !QN)	0.00000	0.00000	0.00000
	(CLK * RN * !SN * !QN)	32.63590	32.63650	32.83130
	(CLK * !RN * QN)	0.00000	0.00000	0.00000
	(CLK * !RN * QN)	31.10080	31.10050	31.10130
	(!CLK * RN * SN)	0.00000	0.00000	0.00000
	(!CLK * RN * SN)	17.57070	17.57400	17.76790
	(!CLK * !RN * QN)	0.00000	0.00000	0.00000
	(!CLK * !RN * QN)	17.19800	17.19880	17.19990

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQNX1	(CLK * RN * SN * QN)	0.00000	0.00000	0.00000
	(CLK * RN * SN * QN)	31.73560	31.73440	31.73450
	(CLK * RN * SN * !QN) + (!CLK * RN * !SN * !QN)	0.00000	0.00000	0.00000
	(CLK * RN * SN * !QN) + (!CLK * RN * !SN * !QN)	15.79800	15.80260	16.01100
	(CLK * RN * !SN * !QN)	0.00000	0.00000	0.00000
	(CLK * RN * !SN * !QN)	15.80290	15.80640	16.02090
	(CLK * !RN * QN)	0.00000	0.00000	0.00000
	(CLK * !RN * QN)	31.72640	31.72630	31.72670
	(!CLK * RN * SN)	0.00000	0.00000	0.00000
	(!CLK * RN * SN)	17.58790	17.59360	17.81060
	(!CLK * !RN * QN)	0.00000	0.00000	0.00000
	(!CLK * !RN * QN)	17.79800	17.79820	17.79810

Passive power(pJ) for RN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQNX1	(CLK * D * SN * QN)	0.00000	0.00000	0.00000
	(CLK * D * SN * QN)	30.36130	30.36060	30.35940
	(CLK * !D * SN * QN)	0.00000	0.00000	0.00000
	(CLK * !D * SN * QN)	30.35870	30.35670	30.35730
	(!CLK * D * SN * QN)	0.00000	0.00000	0.00000
	(!CLK * D * SN * QN)	16.91720	16.91690	17.10540
	(!CLK * !D * SN * QN)	0.00000	0.00000	0.00000
	(!CLK * !D * SN * QN)	16.50690	16.50610	16.50540

Passive power(pJ) for RN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQNX1	(CLK * D * SN * QN)	0.00000	0.00000	0.00000
	(CLK * D * SN * QN)	32.23870	32.24150	32.24160
	(CLK * !D * SN * QN)	0.00000	0.00000	0.00000
	(CLK * !D * SN * QN)	32.23810	32.24180	32.24300
	(!CLK * D * SN * QN)	0.00000	0.00000	0.00000
	(!CLK * D * SN * QN)	18.12250	18.12830	18.32160
	(!CLK * !D * SN * QN)	0.00000	0.00000	0.00000
	(!CLK * !D * SN * QN)	18.30240	18.30060	18.30130

Passive power(pJ) for SN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQNX1	(CLK * D * RN * !QN)	0.00000	0.00000	0.00000
	(CLK * D * RN * !QN)	30.73190	30.74090	30.74120
	(CLK * !RN * QN)	0.00000	0.00000	0.00000
	(CLK * !RN * QN)	32.67410	32.67710	33.06670
	(CLK * !D * RN * !QN)	0.00000	0.00000	0.00000
	(CLK * !D * RN * !QN)	16.85720	16.86510	16.86500
	(!CLK * D * RN * !QN)	0.00000	0.00000	0.00000
	(!CLK * D * RN * !QN)	16.86910	16.86890	16.87160
	(!CLK * !RN * QN)	0.00000	0.00000	0.00000
	(!CLK * !RN * QN)	14.13180	14.13050	14.45800
	(!CLK * !D * RN * !QN)	0.00000	0.00000	0.00000
	(!CLK * !D * RN * !QN)	17.58820	17.58970	17.78140

Passive power(pJ) for SN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQNX1	(CLK * D * RN * !QN)	0.00000	0.00000	0.00000
	(CLK * D * RN * !QN)	31.98990	31.99880	31.99970
	(CLK * !RN * QN)	0.00000	0.00000	0.00000
	(CLK * !RN * QN)	4.96690	4.97763	5.41035
	(CLK * !D * RN * !QN)	0.00000	0.00000	0.00000
	(CLK * !D * RN * !QN)	18.05810	18.06330	18.06340
	(!CLK * D * RN * !QN)	0.00000	0.00000	0.00000
	(!CLK * D * RN * !QN)	18.06250	18.06450	18.06430
	(!CLK * !RN * QN)	0.00000	0.00000	0.00000
	(!CLK * !RN * QN)	0.08692	0.10152	0.55224
	(!CLK * !D * RN * !QN)	0.00000	0.00000	0.00000
	(!CLK * !D * RN * !QN)	6.06938	6.07458	6.28081

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQNX1	(D * RN * SN * !QN)	0.00000	0.00000	0.00000
	(D * RN * SN * !QN)	32.63310	32.63350	32.82490
	(D * RN * !SN * !QN)	0.00000	0.00000	0.00000
	(D * RN * !SN * !QN)	32.63170	32.63200	32.82110
	(!RN * SN * QN)	0.00000	0.00000	0.00000
	(!RN * SN * QN)	32.44750	32.44730	32.63620
	(!RN * !SN * QN)	0.00000	0.00000	0.00000
	(!RN * !SN * QN)	3.97058	3.96957	4.11450
	(!D * RN * SN * QN)	0.00000	0.00000	0.00000
	(!D * RN * SN * QN)	32.45530	32.45590	32.64670
	(!D * RN * !SN * !QN)	0.00000	0.00000	0.00000
	(!D * RN * !SN * !QN)	18.13440	18.14020	18.39260

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQNX1	(D * RN * SN * QN)	0.00000	0.00000	0.00000
	(D * RN * SN * QN)	16.61530	16.61720	16.82800
	(D * RN * SN * !QN)	0.00000	0.00000	0.00000
	(D * RN * SN * !QN)	15.81250	15.81820	16.05840
	(D * RN * !SN * !QN)	0.00000	0.00000	0.00000
	(D * RN * !SN * !QN)	15.80730	15.81870	16.03730
	(!RN * SN * QN)	0.00000	0.00000	0.00000
	(!RN * SN * QN)	16.18290	16.18770	16.38750
	(!RN * !SN * QN)	0.00000	0.00000	0.00000
	(!RN * !SN * QN)	0.08252	0.08713	0.28487
	(!D * RN * SN * QN)	0.00000	0.00000	0.00000
	(!D * RN * SN * QN)	16.19350	16.19900	16.41040
	(!D * RN * SN * !QN)	0.00000	0.00000	0.00000
	(!D * RN * SN * !QN)	17.59690	17.60020	17.84940
	(!D * RN * !SN * !QN)	0.00000	0.00000	0.00000
	(!D * RN * !SN * !QN)	5.63774	5.64264	5.92991

DFFSNRNQX1

sky130_rhbd_tt_1P8_25C.ccs Cell Library:
Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT				OUTPUT
D	RN	SN	CLK	Q
0	1	1	R	0
1	1	1	R	1
x	x	0	x	1
x	0	1	x	0
x	1	1	x	IQ

Footprint

Cell Name	Area
DFFSNRNQX1	231.55840

Pin Capacitance Information

Cell Name	Pin Cap(pf)				Max Cap(pf)
	D	RN	SN	CLK	Q
DFFSNRNQX1	0.01000	0.03363	0.02236	0.02197	3.33800

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFSNRNQX1	0.00000	32.92020	54.41020

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNRNQX1	CLK->Q (RR)	0.16347	0.48256	4.93627
	SN->Q (FR)	0.03880	0.45752	8.26350

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNRNQX1	CLK->Q (RF)	0.24069	0.64613	7.29650
	RN->Q (FF)	0.11781	0.56212	7.94901
	SN->Q (RF)	0.05715	0.51565	9.71233

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNRNQX1	hold	CLK (R)	-0.01083	-0.01454	2.04775
	setup	CLK (R)	0.17912	0.21516	0.71340

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNRNQX1	hold	CLK (R)	-0.06628	-0.12402	-1.05314
	setup	CLK (R)	0.10971	0.17391	2.22704

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQX1	hold	CLK (R)	(RN * SN)	-0.01083	-0.01454	2.04775
	setup	CLK (R)	(RN * SN)	0.17912	0.21516	0.71340

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQX1	hold	CLK (R)	(RN * SN)	-0.06628	-0.12402	-1.05314
	setup	CLK (R)	(RN * SN)	0.10971	0.17391	2.22704

Constraints(ns) for RN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNRNQX1	recovery	CLK (R)	0.16911	0.22713	7.66118
	removal	CLK (R)	-0.01427	-0.02250	-0.06490
	hold	SN (R)	0.03798	0.06251	0.24770
	setup	SN (R)	0.02008	0.03000	1.46419

Constraints(ns) for RN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQX1	recovery	CLK (R)	(D * SN)	0.16911	0.22713	7.66118
	removal	CLK (R)	(D * SN)	-0.01427	-0.02250	-0.06490
	hold	SN (R)	(CLK * D)	0.03703	0.06251	0.24770
	hold	SN (R)	(CLK * !D)	0.03798	0.06251	0.24628
	hold	SN (R)	(!CLK * D)	0.01324	0.03250	0.17964
	hold	SN (R)	(!CLK * !D)	0.01324	0.03250	0.18748
	setup	SN (R)	(CLK * D)	0.00105	0.01750	1.15975
	setup	SN (R)	(CLK * !D)	0.00258	0.01750	1.46419
	setup	SN (R)	(!CLK * D)	0.02008	0.03000	1.00834
	setup	SN (R)	(!CLK * !D)	0.01967	0.03000	0.85039

Constraints(ns) for RN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQX1	min_pulse_width	RN ()	(CLK * D * SN)	0.11249	0.47852	16.50020
	min_pulse_width	RN ()	(CLK * !D * SN)	0.10162	0.47852	16.50020
	min_pulse_width	RN ()	(!CLK * D * SN)	0.06538	0.47852	16.50020
	min_pulse_width	RN ()	(!CLK * !D * SN)	0.06538	0.47852	16.50020

Constraints(ns) for SN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNRNQX1	recovery	CLK (R)	0.02768	0.01789	4.05542
	removal	CLK (R)	0.00269	0.00750	0.01365
	hold	RN (R)	0.04988	0.10251	0.66541
	setup	RN (R)	0.01277	0.00000	0.61219

Constraints(ns) for SN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQX1	recovery	CLK (R)	(!D * RN)	0.02768	0.01789	4.05542
	removal	CLK (R)	(!D * RN)	0.00269	0.00750	0.01365
	hold	RN (R)	CLK	0.04399	0.10251	0.66541
	hold	RN (R)	(!CLK * D)	0.04988	0.10251	0.60462
	hold	RN (R)	(!CLK * !D)	0.04838	0.10001	0.63165
	setup	RN (R)	CLK	0.01277	0.00000	0.61219
	setup	RN (R)	(!CLK * D)	-0.01506	-0.02000	0.46509
	setup	RN (R)	(!CLK * !D)	-0.01346	-0.02250	0.09471

Constraints(ns) for SN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQX1	min_pulse_width	SN ()	(CLK * D * RN)	0.08713	0.47852	16.50020
	min_pulse_width	SN ()	(CLK * !D * RN)	0.08713	0.47852	16.50020
	min_pulse_width	SN ()	(!CLK * D * RN)	0.07263	0.47852	16.50020
	min_pulse_width	SN ()	(!CLK * !D * RN)	0.07263	0.47852	16.50020

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQX1	min_pulse_width	CLK ()	(D * RN * SN)	0.15961	0.47852	16.50020
	min_pulse_width	CLK ()	(!D * RN * SN)	0.14874	0.47852	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQX1	min_pulse_width	CLK ()	(D * RN * SN)	0.24296	0.47852	16.50020
	min_pulse_width	CLK ()	(!D * RN * SN)	0.11974	0.47852	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNRNQX1	CLK	0.00000	0.00000	0.00000
	CLK	32.67020	32.67220	32.93830
	SN	32.59430	32.60020	32.84690

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNRNQX1	CLK	0.00000	0.00000	0.00000
	CLK	32.50480	32.50810	32.70790
	RN	-0.00228	-0.24133	-5.40752
	RN	33.05890	33.07220	33.41320
	SN	32.67530	32.67640	32.87420

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQX1	(CLK * RN * SN * Q) + (!CLK * RN * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * RN * SN * Q) + (!CLK * RN * !SN * Q)	32.63690	32.63790	32.83090
	(CLK * RN * SN * !Q)	0.00000	0.00000	0.00000
	(CLK * RN * SN * !Q)	31.10890	31.10890	31.10960
	(CLK * RN * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * RN * !SN * Q)	32.63580	32.63650	32.83130
	(CLK * !RN * SN * !Q) + (CLK * !RN * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * !RN * SN * !Q) + (CLK * !RN * !SN * Q)	31.10080	31.10040	31.10140
	(!CLK * RN * SN)	0.00000	0.00000	0.00000
	(!CLK * RN * SN)	17.56780	17.56880	17.76100
	(!CLK * !RN * SN * !Q) + (!CLK * !RN * !SN * Q)	0.00000	0.00000	0.00000
	(!CLK * !RN * SN * !Q) + (!CLK * !RN * !SN * Q)	17.19810	17.19890	17.19990

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQX1	(CLK * RN * SN * Q) + (!CLK * RN * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * RN * SN * Q) + (!CLK * RN * !SN * Q)	15.79780	15.80250	16.01100
	(CLK * RN * SN * !Q)	0.00000	0.00000	0.00000
	(CLK * RN * SN * !Q)	31.73560	31.73440	31.73450
	(CLK * RN * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * RN * !SN * Q)	15.80280	15.80630	16.02080
	(CLK * !RN * SN * !Q) + (CLK * !RN * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * !RN * SN * !Q) + (CLK * !RN * !SN * Q)	31.72650	31.72640	31.72680
	(!CLK * RN * SN)	0.00000	0.00000	0.00000
	(!CLK * RN * SN)	17.58320	17.58920	17.80170
	(!CLK * !RN * SN * !Q) + (!CLK * !RN * !SN * Q)	0.00000	0.00000	0.00000
	(!CLK * !RN * SN * !Q) + (!CLK * !RN * !SN * Q)	17.79810	17.79820	17.79810

Passive power(pJ) for RN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQX1	(CLK * D * SN * !Q)	0.00000	0.00000	0.00000
	(CLK * D * SN * !Q)	30.36280	30.36170	30.36070
	(CLK * D * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * D * !SN * Q)	32.88690	32.88670	33.26110
	(CLK * !D * SN * !Q)	0.00000	0.00000	0.00000
	(CLK * !D * SN * !Q)	30.35840	30.35650	30.35700
	(CLK * !D * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * !D * !SN * Q)	17.62490	17.62290	17.90200
	(!CLK * D * SN * !Q)	0.00000	0.00000	0.00000
	(!CLK * D * SN * !Q)	16.91740	16.91690	17.10540
	(!CLK * D * !SN * Q)	0.00000	0.00000	0.00000
	(!CLK * D * !SN * Q)	14.12730	14.12670	14.46370
	(!CLK * !D * SN * !Q)	0.00000	0.00000	0.00000
	(!CLK * !D * SN * !Q)	16.50700	16.50620	16.50550
	(!CLK * !D * !SN * Q)	0.00000	0.00000	0.00000
	(!CLK * !D * !SN * Q)	3.61099	3.61100	3.77756

Passive power(pJ) for RN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQX1	(CLK * D * SN * !Q)	0.00000	0.00000	0.00000
	(CLK * D * SN * !Q)	32.23840	32.24140	32.24140
	(CLK * D * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * D * !SN * Q)	4.83801	4.85010	5.44802
	(CLK * !D * SN * !Q)	0.00000	0.00000	0.00000
	(CLK * !D * SN * !Q)	32.23960	32.24220	32.24300
	(CLK * !D * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * !D * !SN * Q)	6.13290	6.13784	6.53271
	(!CLK * D * SN * !Q)	0.00000	0.00000	0.00000
	(!CLK * D * SN * !Q)	18.12280	18.12820	18.32170
	(!CLK * D * !SN * Q)	0.00000	0.00000	0.00000
	(!CLK * D * !SN * Q)	0.10811	0.12123	0.55290
	(!CLK * !D * SN * !Q)	0.00000	0.00000	0.00000
	(!CLK * !D * SN * !Q)	18.30290	18.30070	18.30130
	(!CLK * !D * !SN * Q)	0.00000	0.00000	0.00000
	(!CLK * !D * !SN * Q)	0.08231	0.08867	0.27864

Passive power(pJ) for SN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQX1	(CLK * D * RN * Q)	0.00000	0.00000	0.00000
	(CLK * D * RN * Q)	30.73240	30.73980	30.73980
	(CLK * !D * RN * Q)	0.00000	0.00000	0.00000
	(CLK * !D * RN * Q)	16.86010	16.86440	16.86520
	(!CLK * D * RN * Q)	0.00000	0.00000	0.00000
	(!CLK * D * RN * Q)	16.87100	16.86870	16.87150
	(!CLK * !D * RN * Q)	0.00000	0.00000	0.00000
	(!CLK * !D * RN * Q)	17.58770	17.58930	17.78090

Passive power(pJ) for SN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQX1	(CLK * D * RN * Q)	0.00000	0.00000	0.00000
	(CLK * D * RN * Q)	31.99140	31.99840	31.99950
	(CLK * !D * RN * Q)	0.00000	0.00000	0.00000
	(CLK * !D * RN * Q)	18.05960	18.06350	18.06360
	(!CLK * D * RN * Q)	0.00000	0.00000	0.00000
	(!CLK * D * RN * Q)	18.06240	18.06430	18.06420
	(!CLK * !D * RN * Q)	0.00000	0.00000	0.00000
	(!CLK * !D * RN * Q)	6.06944	6.07479	6.28113

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQX1	(D * RN * SN * Q)	0.00000	0.00000	0.00000
	(D * RN * SN * Q)	32.63340	32.63380	32.82470
	(D * RN * !SN * Q)	0.00000	0.00000	0.00000
	(D * RN * !SN * Q)	32.63300	32.63330	32.82240
	(!RN * SN * !Q)	0.00000	0.00000	0.00000
	(!RN * SN * !Q)	32.44650	32.44630	32.63520
	(!RN * !SN * Q)	0.00000	0.00000	0.00000
	(!RN * !SN * Q)	3.97069	3.96968	4.11450
	(!D * RN * SN * !Q)	0.00000	0.00000	0.00000
	(!D * RN * SN * !Q)	32.45550	32.45610	32.64690
	(!D * RN * !SN * Q)	0.00000	0.00000	0.00000
	(!D * RN * !SN * Q)	18.13450	18.14000	18.39260

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQX1	(D * RN * SN * Q)	0.00000	0.00000	0.00000
	(D * RN * SN * Q)	15.81090	15.81910	16.05910
	(D * RN * SN * !Q)	0.00000	0.00000	0.00000
	(D * RN * SN * !Q)	16.61730	16.61880	16.82950
	(D * RN * !SN * Q)	0.00000	0.00000	0.00000
	(D * RN * !SN * Q)	15.81000	15.81850	16.03750
	(!RN * SN * !Q)	0.00000	0.00000	0.00000
	(!RN * SN * !Q)	16.18240	16.18690	16.38690
	(!RN * !SN * Q)	0.00000	0.00000	0.00000
	(!RN * !SN * Q)	0.08252	0.08710	0.28488
	(!D * RN * SN * Q)	0.00000	0.00000	0.00000
	(!D * RN * SN * Q)	17.59780	17.60000	17.84930
	(!D * RN * SN * !Q)	0.00000	0.00000	0.00000
	(!D * RN * SN * !Q)	16.19440	16.19600	16.40910
	(!D * RN * !SN * Q)	0.00000	0.00000	0.00000
	(!D * RN * !SN * Q)	5.63719	5.64236	5.92965

DFFSNRNX1

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Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT				OUTPUT	
D	**RN**	**SN**	**CLK**	**Q**	**QN**
0	1	1	R	0	1
1	1	1	R	1	0
x	0	0	x	1	1
x	0	1	x	0	1
x	1	0	x	1	0
x	1	1	x	IQ	IQN

Footprint

Cell Name	Area
DFFSNRNX1	231.55840

Pin Capacitance Information

Cell Name	Pin Cap(pf)				Max Cap(pf)	
	D	**RN**	**SN**	**CLK**	**Q**	**QN**
DFFSNRNX1	0.01000	0.03392	0.02236	0.02197	3.33791	3.37818

Leakage Information

Cell Name	Leakage(nW)		
	Min.	**Avg**	**Max.**
DFFSNRNX1	0.00000	32.92020	54.41020

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNRNX1	CLK->Q (RR)	0.16371	0.48227	4.93937
	QN->Q (FR)	0.04822	0.45793	8.12695
	SN->Q (FR)	0.03884	0.45693	8.27386

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNRNX1	CLK->Q (RF)	0.24250	0.90987	13.26950
	QN->Q (RF)	0.07172	0.47431	8.63798
	RN->Q (FF)	0.11905	0.91412	16.74560
	SN->Q (RF)	0.05721	0.51563	9.71211

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNRNX1	CLK->QN (RR)	0.16032	0.46224	5.01046
	Q->QN (FR)	0.03563	0.44141	8.07078
	RN->QN (FR)	0.03875	0.45732	8.32274

Delay(ns) to QN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNRNX1	CLK->QN (RF)	0.23423	0.97960	14.73290
	Q->QN (RF)	0.05210	0.52055	9.70176
	RN->QN (RF)	0.26063	0.71529	10.20950
	SN->QN (FF)	0.10575	0.93474	17.71220

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNRNX1	hold	CLK (R)	-0.01083	-0.01452	2.02825
	setup	CLK (R)	0.17874	0.21499	0.71348

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNRNX1	hold	CLK (R)	-0.06599	-0.12402	-1.05291
	setup	CLK (R)	0.10977	0.17343	2.19055

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNX1	hold	CLK (R)	(RN * SN)	-0.01083	-0.01452	2.02825
	setup	CLK (R)	(RN * SN)	0.17874	0.21499	0.71348

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNX1	hold	CLK (R)	(RN * SN)	-0.06599	-0.12402	-1.05291
	setup	CLK (R)	(RN * SN)	0.10977	0.17343	2.19055

Constraints(ns) for RN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNRNX1	recovery	CLK (R)	0.17111	0.22720	7.63667
	removal	CLK (R)	-0.01427	-0.02250	-0.06490
	hold	SN (R)	0.02381	0.05501	0.44272
	setup	SN (R)	0.02136	0.03400	1.06633

Constraints(ns) for RN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNX1	recovery	CLK (R)	(D * SN)	0.17111	0.22720	7.63667
	removal	CLK (R)	(D * SN)	-0.01427	-0.02250	-0.06490
	hold	SN (R)	(CLK * D)	0.00539	-0.01250	-0.10673
	hold	SN (R)	(CLK * !D)	0.00387	-0.01250	-0.11327
	hold	SN (R)	(!CLK * D)	0.02229	0.05501	0.43139
	hold	SN (R)	(!CLK * !D)	0.02381	0.05501	0.44272
	setup	SN (R)	(CLK * D)	0.00898	0.02400	0.54104
	setup	SN (R)	(CLK * !D)	0.00826	0.02400	0.64146
	setup	SN (R)	(!CLK * D)	0.02136	0.03400	1.06633
	setup	SN (R)	(!CLK * !D)	0.02093	0.03400	0.91672

Constraints(ns) for RN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNX1	min_pulse_width	RN ()	(CLK * D * SN)	0.11249	0.47852	16.50020
	min_pulse_width	RN ()	(CLK * !D * SN)	0.10162	0.47852	16.50020
	min_pulse_width	RN ()	(!CLK * D * SN)	0.06901	0.47852	16.50020
	min_pulse_width	RN ()	(!CLK * !D * SN)	0.06901	0.47852	16.50020

Constraints(ns) for SN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNRNX1	recovery	CLK (R)	0.02718	0.01799	3.94022
	removal	CLK (R)	0.00269	0.00750	0.01314
	hold	RN (R)	0.05079	0.10501	0.66541
	setup	RN (R)	0.01275	0.00000	0.60876

Constraints(ns) for SN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNX1	recovery	CLK (R)	(!D * RN)	0.02718	0.01799	3.94022
	removal	CLK (R)	(!D * RN)	0.00269	0.00750	0.01314
	hold	RN (R)	CLK	0.04399	0.10251	0.66541
	hold	RN (R)	(!CLK * D)	0.04832	0.10501	0.62936
	hold	RN (R)	(!CLK * !D)	0.05079	0.10501	0.64202
	setup	RN (R)	CLK	0.01275	0.00000	0.60876
	setup	RN (R)	(!CLK * D)	-0.01577	-0.02500	0.45738
	setup	RN (R)	(!CLK * !D)	-0.01577	-0.02500	0.06792

Constraints(ns) for SN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNX1	min_pulse_width	SN ()	(CLK * D * RN)	0.08713	0.47852	16.50020
	min_pulse_width	SN ()	(CLK * !D * RN)	0.08713	0.47852	16.50020
	min_pulse_width	SN ()	(!CLK * D * RN)	0.07626	0.47852	16.50020
	min_pulse_width	SN ()	(!CLK * !D * RN)	0.07626	0.47852	16.50020

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNX1	min_pulse_width	CLK ()	(D * RN * SN)	0.16323	0.47852	16.50020
	min_pulse_width	CLK ()	(!D * RN * SN)	0.14874	0.47852	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNX1	min_pulse_width	CLK ()	(D * RN * SN)	0.24296	0.47852	16.50020
	min_pulse_width	CLK ()	(!D * RN * SN)	0.11974	0.47852	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNRNX1	CLK	0.00000	0.00000	0.00000
	CLK	16.33510	16.33570	16.42970
	SN	-0.00114	-0.12066	-2.70369
	SN	16.29730	16.30100	16.39900

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNRNX1	CLK	0.00000	0.00000	0.00000
	CLK	16.25240	16.25340	16.35370
	RN	-0.00114	-0.12066	-2.70369
	RN	16.52940	16.53350	16.63690
	SN	32.67530	32.67650	32.87420

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNRNX1	CLK	0.00000	0.00000	0.00000
	CLK	16.25200	16.25300	16.35390
	RN	-0.00114	-0.12154	-2.73630
	RN	16.52970	16.53410	16.63650

Internal switching power(pJ) to QN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNRNX1	CLK	0.00000	0.00000	0.00000
	CLK	16.33540	16.33600	16.43040
	RN	32.88890	32.89100	33.16220
	SN	-0.00114	-0.12154	-2.73630
	SN	16.29740	16.30110	16.39620

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNX1	(CLK * RN * SN * Q * !QN) + (!CLK * RN * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * RN * SN * Q * !QN) + (!CLK * RN * !SN * Q * !QN)	32.63690	32.63790	32.83090
	(CLK * RN * SN * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * RN * SN * !Q * QN)	31.10860	31.10890	31.10960
	(CLK * RN * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * RN * !SN * Q * !QN)	32.63590	32.63650	32.83130
	(CLK * !RN * SN * !Q * QN) + (CLK * !RN * !SN * Q * QN)	0.00000	0.00000	0.00000
	(CLK * !RN * SN * !Q * QN) + (CLK * !RN * !SN * Q * QN)	31.10080	31.10050	31.10140
	(!CLK * RN * SN * Q * !QN) + (!CLK * RN * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * RN * SN * Q * !QN) + (!CLK * RN * SN * !Q * QN)	17.57070	17.57350	17.76790
	(!CLK * !RN * SN * !Q * QN) + (!CLK * !RN * !SN * Q * QN)	0.00000	0.00000	0.00000
	(!CLK * !RN * SN * !Q * QN) + (!CLK * !RN * !SN * Q * QN)	17.19810	17.19890	17.19990

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNX1	(CLK * RN * SN * Q * !QN) + (!CLK * RN * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * RN * SN * Q * !QN) + (!CLK * RN * !SN * Q * !QN)	15.79790	15.80260	16.01100
	(CLK * RN * SN * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * RN * SN * !Q * QN)	31.73560	31.73440	31.73450
	(CLK * RN * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * RN * !SN * Q * !QN)	15.80290	15.80630	16.02080
	(CLK * !RN * SN * !Q * QN) + (CLK * !RN * !SN * Q * QN)	0.00000	0.00000	0.00000
	(CLK * !RN * SN * !Q * QN) + (CLK * !RN * !SN * Q * QN)	31.72640	31.72640	31.72680
	(!CLK * RN * SN * Q * !QN) + (!CLK * RN * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * RN * SN * Q * !QN) + (!CLK * RN * SN * !Q * QN)	17.58790	17.59360	17.81060
	(!CLK * !RN * SN * !Q * QN) + (!CLK * !RN * !SN * Q * QN)	0.00000	0.00000	0.00000
	(!CLK * !RN * SN * !Q * QN) + (!CLK * !RN * !SN * Q * QN)	17.79810	17.79820	17.79810

Passive power(pJ) for RN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNX1	(CLK * D * SN * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * D * SN * !Q * QN)	30.36140	30.36070	30.35950
	(CLK * !D * SN * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * !D * SN * !Q * QN)	30.35870	30.35680	30.35740
	(!CLK * D * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * D * SN * !Q * QN)	16.91730	16.91690	17.10540
	(!CLK * !D * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * !D * SN * !Q * QN)	16.50690	16.50610	16.50550

Passive power(pJ) for RN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNX1	(CLK * D * SN * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * D * SN * !Q * QN)	32.23900	32.24160	32.24170
	(CLK * !D * SN * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * !D * SN * !Q * QN)	32.23670	32.24200	32.24300
	(!CLK * D * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * D * SN * !Q * QN)	18.12250	18.12880	18.32170
	(!CLK * !D * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * !D * SN * !Q * QN)	18.30260	18.30060	18.30130

Passive power(pJ) for SN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNX1	(CLK * D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * D * RN * Q * !QN)	30.73250	30.73980	30.73980
	(CLK * !D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * !D * RN * Q * !QN)	16.85890	16.86570	16.86560
	(!CLK * D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(!CLK * D * RN * Q * !QN)	16.87120	16.86880	16.87160
	(!CLK * !D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(!CLK * !D * RN * Q * !QN)	17.58850	17.58980	17.78140

Passive power(pJ) for SN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNX1	(CLK * D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * D * RN * Q * !QN)	31.99110	31.99800	31.99910
	(CLK * !D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * !D * RN * Q * !QN)	18.05970	18.06350	18.06360
	(!CLK * D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(!CLK * D * RN * Q * !QN)	18.06250	18.06440	18.06430
	(!CLK * !D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(!CLK * !D * RN * Q * !QN)	6.06962	6.07502	6.28123

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNX1	(D * RN * SN * Q * !QN)	0.00000	0.00000	0.00000
	(D * RN * SN * Q * !QN)	32.63310	32.63360	32.82480
	(D * RN * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(D * RN * !SN * Q * !QN)	32.63190	32.63220	32.82130
	(!RN * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!RN * SN * !Q * QN)	32.44750	32.44730	32.63630
	(!RN * !SN * Q * QN)	0.00000	0.00000	0.00000
	(!RN * !SN * Q * QN)	3.97058	3.96957	4.11449
	(!D * RN * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!D * RN * SN * !Q * QN)	32.45530	32.45590	32.64670
	(!D * RN * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(!D * RN * !SN * Q * !QN)	18.13440	18.14000	18.39260

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNX1	(D * RN * SN * Q * !QN)	0.00000	0.00000	0.00000
	(D * RN * SN * Q * !QN)	15.81330	15.81800	16.05840
	(D * RN * SN * !Q * QN)	0.00000	0.00000	0.00000
	(D * RN * SN * !Q * QN)	16.61530	16.61720	16.82800
	(D * RN * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(D * RN * !SN * Q * !QN)	15.80730	15.81830	16.03730
	(!RN * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!RN * SN * !Q * QN)	16.18540	16.18940	16.38930
	(!RN * !SN * Q * QN)	0.00000	0.00000	0.00000
	(!RN * !SN * Q * QN)	0.08252	0.08713	0.28487
	(!D * RN * SN * Q * !QN)	0.00000	0.00000	0.00000
	(!D * RN * SN * Q * !QN)	17.59750	17.60000	17.84940
	(!D * RN * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!D * RN * SN * !Q * QN)	16.19370	16.19900	16.41040
	(!D * RN * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(!D * RN * !SN * Q * !QN)	5.63641	5.64256	5.92990

DFFSNX1

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Truth Table

INPUT			OUTPUT	
D	**SN**	**CLK**	**Q**	**QN**
0	1	R	0	1
1	1	R	1	0
x	0	x	1	0
x	1	x	IQ	IQN

Footprint

Cell Name	Area
DFFSNX1	197.10400

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)	
	D	**SN**	**CLK**	**Q**	**QN**
DFFSNX1	0.01000	0.02129	0.02197	3.46739	4.79830

Leakage Information

Cell Name	Leakage(nW)		
	Min.	**Avg**	**Max.**
DFFSNX1	0.00000	30.10450	57.62650

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNX1	CLK->Q (RR)	0.11666	0.43263	4.90991
	QN->Q (FR)	0.04761	0.46262	8.28782
	SN->Q (FR)	0.03837	0.46066	8.43448

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNX1	CLK->Q (RF)	0.22730	0.90897	13.75810
	QN->Q (RF)	0.07090	0.48251	8.93144

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNX1	CLK->QN (RR)	0.14622	0.48920	6.65409
	Q->QN (FR)	0.03222	0.48187	9.70840

Delay(ns) to QN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNX1	CLK->QN (RF)	0.16760	0.93797	17.35450
	Q->QN (RF)	0.03481	0.47396	9.83753
	SN->QN (FF)	0.08865	0.95479	20.66710

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNX1	hold	CLK (R)	0.01290	0.01557	0.78620
	setup	CLK (R)	0.14297	0.19260	3.34493

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNX1	hold	CLK (R)	-0.07203	-0.13252	-1.11443
	setup	CLK (R)	0.10428	0.16901	1.57496

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNX1	hold	CLK (R)	SN	0.01290	0.01557	0.78620
	setup	CLK (R)	SN	0.14297	0.19260	3.34493

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNX1	hold	CLK (R)	SN	-0.07203	-0.13252	-1.11443
	setup	CLK (R)	SN	0.10428	0.16901	1.57496

Constraints(ns) for SN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNX1	recovery	CLK (R)	0.03447	0.03601	4.22813
	removal	CLK (R)	-0.01435	-0.01500	-0.11230

Constraints(ns) for SN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNX1	recovery	CLK (R)	!D	0.03447	0.03601	4.22813
	removal	CLK (R)	!D	-0.01435	-0.01500	-0.11230

Constraints(ns) for SN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNX1	min_pulse_width	SN ()	(CLK * D)	0.07263	0.47852	16.50020
	min_pulse_width	SN ()	(CLK * !D)	0.06901	0.47852	16.50020
	min_pulse_width	SN ()	(!CLK * D)	0.06538	0.47852	16.50020
	min_pulse_width	SN ()	(!CLK * !D)	0.06538	0.47852	16.50020

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNX1	min_pulse_width	CLK ()	(D * SN)	0.10887	0.47852	16.50020
	min_pulse_width	CLK ()	(!D * SN)	0.14149	0.47852	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNX1	min_pulse_width	CLK ()	(D * SN)	0.21759	0.47852	16.50020
	min_pulse_width	CLK ()	(!D * SN)	0.10887	0.47852	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNX1	CLK	0.00000	0.00000	0.00000
	CLK	9.81940	9.82127	9.93215
	SN	-0.00114	-0.12345	-2.80858
	SN	9.49785	9.50266	9.60211

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNX1	CLK	0.00000	0.00000	0.00000
	CLK	17.68060	17.68210	17.78420

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNX1	CLK	0.00000	0.00000	0.00000
	CLK	17.68060	17.68180	17.78460

Internal switching power(pJ) to QN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNX1	CLK	0.00000	0.00000	0.00000
	CLK	9.81966	9.82092	9.93265
	SN	-0.00114	-0.15002	-3.88661
	SN	9.49752	9.50188	9.59770

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNX1	(CLK * SN * Q * !QN) + (!CLK * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * SN * Q * !QN) + (!CLK * !SN * Q * !QN)	19.80270	19.80400	20.00250
	(CLK * SN * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * SN * !Q * QN)	33.71210	33.71320	33.71330
	(CLK * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * !SN * Q * !QN)	19.79880	19.80010	20.00030
	(!CLK * SN * Q * !QN) + (!CLK * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * SN * Q * !QN) + (!CLK * SN * !Q * QN)	10.28500	10.28710	10.48710

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNX1	(CLK * SN * Q * !QN) + (!CLK * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * SN * Q * !QN) + (!CLK * !SN * Q * !QN)	9.55581	9.55972	9.76764
	(CLK * SN * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * SN * !Q * QN)	34.35950	34.35970	34.35960
	(CLK * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * !SN * Q * !QN)	9.55900	9.56256	9.77833
	(!CLK * SN * Q * !QN) + (!CLK * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * SN * Q * !QN) + (!CLK * SN * !Q * QN)	14.91880	14.92560	15.13550

Passive power(pJ) for SN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNX1	(CLK * D * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * D * Q * !QN)	18.32360	18.33090	18.33220
	(CLK * !D * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * !D * Q * !QN)	9.86628	9.87297	9.87273
	(!CLK * D * Q * !QN)	0.00000	0.00000	0.00000
	(!CLK * D * Q * !QN)	9.88020	9.87604	9.88006
	(!CLK * !D * Q * !QN)	0.00000	0.00000	0.00000
	(!CLK * !D * Q * !QN)	14.40920	14.41060	14.60110

Passive power(pJ) for SN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNX1	(CLK * D * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * D * Q * !QN)	19.54910	19.55490	19.55600
	(CLK * !D * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * !D * Q * !QN)	11.02310	11.02650	11.02690
	(!CLK * D * Q * !QN)	0.00000	0.00000	0.00000
	(!CLK * D * Q * !QN)	11.02720	11.02730	11.02730
	(!CLK * !D * Q * !QN)	0.00000	0.00000	0.00000
	(!CLK * !D * Q * !QN)	3.67471	3.68056	3.88712

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNX1	(D * SN * Q * !QN)	0.00000	0.00000	0.00000
	(D * SN * Q * !QN)	19.79580	19.79870	20.01600
	(D * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(D * !SN * Q * !QN)	19.79260	19.79380	20.00860
	(!D * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!D * SN * !Q * QN)	35.14030	35.13980	35.33860
	(!D * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(!D * !SN * Q * !QN)	10.63750	10.64740	10.90790

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNX1	(D * SN * Q * !QN)	0.00000	0.00000	0.00000
	(D * SN * Q * !QN)	9.56807	9.57901	9.84267
	(D * SN * !Q * QN)	0.00000	0.00000	0.00000
	(D * SN * !Q * QN)	13.00200	13.00060	13.24120
	(D * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(D * !SN * Q * !QN)	9.55928	9.57126	9.81205
	(!D * SN * Q * !QN)	0.00000	0.00000	0.00000
	(!D * SN * Q * !QN)	14.92570	14.93150	15.17550
	(!D * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!D * SN * !Q * QN)	17.39180	17.39240	17.60400
	(!D * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(!D * !SN * Q * !QN)	3.48900	3.49616	3.79537

DFFX1

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Truth Table

INPUT		OUTPUT	
D	CLK	Q	QN
0	R	0	1
1	R	1	0
x	x	IQ	IQN

Footprint

Cell Name	Area
DFFX1	174.13440

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)	
	D	CLK	Q	QN
DFFX1	0.01036	0.02277	5.00848	5.02235

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFX1	0.00000	28.46890	43.32350

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFX1	CLK->Q (RR)	0.12148	0.49401	6.81902
	QN->Q (FR)	0.03898	0.49895	10.10790

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFX1	CLK->Q (RF)	0.20414	0.93544	17.04540
	QN->Q (RF)	0.04435	0.46152	9.75418

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFX1	CLK->QN (RR)	0.14853	0.49813	6.90478
	Q->QN (FR)	0.03259	0.48833	10.01110

Delay(ns) to QN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFX1	CLK->QN (RF)	0.17244	0.97100	18.14390
	Q->QN (RF)	0.03565	0.48742	10.26900

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFX1	hold	CLK (R)	0.04761	0.07601	0.71629
	setup	CLK (R)	0.11508	0.16461	0.79796

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFX1	hold	CLK (R)	-0.03408	-0.07663	-0.69366
	setup	CLK (R)	0.06159	0.10940	1.49281

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFX1	min_pulse_width	CLK ()	D	0.11249	0.47852	16.50020
	min_pulse_width	CLK ()	!D	0.13786	0.47852	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFX1	min_pulse_width	CLK ()	D	0.20309	0.47852	16.50020
	min_pulse_width	CLK ()	!D	0.09438	0.47852	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFX1	CLK	0.00000	0.00000	0.00000
	CLK	10.56480	10.56590	10.67190

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFX1	CLK	0.00000	0.00000	0.00000
	CLK	13.12780	13.12950	13.23400

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFX1	CLK	0.00000	0.00000	0.00000
	CLK	13.12780	13.12940	13.23340

Internal switching power(pJ) to QN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFX1	CLK	0.00000	0.00000	0.00000
	CLK	10.56480	10.56580	10.67330

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFX1	(CLK * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * Q * !QN)	21.09460	21.09740	21.31170
	(CLK * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * !Q * QN)	24.93340	24.93590	24.93770
	(!CLK * Q * !QN) + (!CLK * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * Q * !QN) + (!CLK * !Q * QN)	11.29700	11.29990	11.51240

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFX1	(CLK * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * Q * !QN)	10.10230	10.10810	10.33990
	(CLK * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * !Q * QN)	25.56850	25.56860	25.56870
	(!CLK * Q * !QN) + (!CLK * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * Q * !QN) + (!CLK * !Q * QN)	11.30920	11.31500	11.53250

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFX1	(D * Q * !QN)	0.00000	0.00000	0.00000
	(D * Q * !QN)	21.09420	21.09480	21.30240
	(!D * !Q * QN)	0.00000	0.00000	0.00000
	(!D * !Q * QN)	26.21450	26.21520	26.41230

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFX1	(D * Q * !QN)	0.00000	0.00000	0.00000
	(D * Q * !QN)	10.11810	10.11850	10.34480
	(D * !Q * QN)	0.00000	0.00000	0.00000
	(D * !Q * QN)	10.32780	10.32970	10.53830
	(!D * Q * !QN)	0.00000	0.00000	0.00000
	(!D * Q * !QN)	11.32260	11.32600	11.54160
	(!D * !Q * QN)	0.00000	0.00000	0.00000
	(!D * !Q * QN)	9.90926	9.91236	10.11490

DLATCHN

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Truth Table

INPUT		OUTPUT
D	GATE_N	Q
0	0	0
x	1	1
1	x	1

Footprint

Cell Name	Area
DLATCHN	179.87680

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	D	GATE_N	Q
DLATCHN	0.02247	0.01029	2.72300

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DLATCHN	0.00000	22.21230	28.71050

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DLATCHN	D->Q (RR)	0.17217	0.57771	6.90796
	GATE_N->Q (-R)	0.22271	0.68899	7.91568

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DLATCHN	D->Q (FF)	0.15501	0.38550	3.31539
	GATE_N->Q (-F)	0.17035	0.42666	3.40660

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DLATCHN	D	0.00000	0.00000	0.00000
	D	13.54220	13.55700	14.08890
	GATE_N	0.00000	0.00000	0.00000
	GATE_N	13.29170	13.30010	13.53620

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DLATCHN	D	0.00000	0.00000	0.00000
	D	9.19352	9.20877	9.72956
	GATE_N	0.00000	0.00000	0.00000
	GATE_N	7.78817	7.79172	8.08091

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DLATCHN	GATE_N	0.00000	0.00000	0.00000
	GATE_N	16.22070	16.22760	16.46980

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DLATCHN	GATE_N	0.00000	0.00000	0.00000
	GATE_N	11.66250	11.67190	11.90740

Passive power(pJ) for GATE_N rising (conditional):

Cell Name	**When**	**Power(pJ)**		
		first	**mid**	**last**
DLATCHN	**(D * Q)**	0.00000	0.00000	0.00000
	(D * Q)	16.31760	16.32330	16.59260
	(!D * !Q)	0.00000	0.00000	0.00000
	(!D * !Q)	11.95510	11.96140	12.20090

Passive power(pJ) for GATE_N falling (conditional):

Cell Name	**When**	**Power(pJ)**		
		first	**mid**	**last**
DLATCHN	**(D * Q)**	0.00000	0.00000	0.00000
	(D * Q)	11.76870	11.77750	12.03430
	(!D * !Q)	0.00000	0.00000	0.00000
	(!D * !Q)	8.99221	9.00029	9.24848

DLATCH

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Truth Table

INPUT		OUTPUT
D	**GATE**	**Q**
x	0	1
0	1	0
1	1	1

Footprint

Cell Name	Area
DLATCH	162.64960

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	D	**GATE**	**Q**
DLATCH	0.02249	0.02038	2.76163

Leakage Information

Cell Name	Leakage(nW)		
	Min.	**Avg**	**Max.**
DLATCH	0.00000	18.14480	24.27300

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DLATCH	D->Q (RR)	0.17169	0.57844	6.96902
	GATE->Q (-R)	0.17123	0.56812	6.88450

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DLATCH	D->Q (FF)	0.15311	0.38429	3.33989
	GATE->Q (-F)	0.11275	0.31251	2.57521

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DLATCH	D	0.00000	0.00000	0.00000
	D	14.12600	14.13950	14.68310
	GATE	0.00000	0.00000	0.00000
	GATE	13.88640	13.89070	14.14150

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DLATCH	D	0.00000	0.00000	0.00000
	D	9.55120	9.56744	10.09200
	GATE	0.00000	0.00000	0.00000
	GATE	8.12073	8.12095	8.40042

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DLATCH	!GATE	0.00000	0.00000	0.00000
	!GATE	12.15680	12.16330	12.40560

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DLATCH	!GATE	0.00000	0.00000	0.00000
	!GATE	8.01316	8.02238	8.25679

Passive power(pJ) for GATE rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DLATCH	(D * Q)	0.00000	0.00000	0.00000
	(D * Q)	12.37180	12.37660	12.60300
	(!D * !Q)	0.00000	0.00000	0.00000
	(!D * !Q)	9.32356	9.32993	9.54953

Passive power(pJ) for GATE falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DLATCH	(D * Q)	0.00000	0.00000	0.00000
	(D * Q)	12.18710	12.19620	12.45060
	(!D * !Q)	0.00000	0.00000	0.00000
	(!D * !Q)	8.23656	8.24292	8.47074

FA

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Truth Table

INPUT			OUTPUT	
A	B	CIN	COUT	SUM
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

Footprint

Cell Name	Area
FA	309.08081

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)	
	A	B	CIN	COUT	SUM
FA	0.03702	0.03673	0.03414	5.57833	2.41930

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
FA	0.00000	36.98210	51.95560

Delay Information

Delay(ns) to COUT rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
FA	A->COUT (RR)	0.32381	0.62894	6.92479
	B->COUT (RR)	0.34934	0.65699	6.98752
	CIN->COUT (RR)	0.14110	0.45667	6.84245

Delay(ns) to COUT falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
FA	A->COUT (FF)	0.29718	0.60997	5.93051
	B->COUT (FF)	0.30706	0.62132	6.01443
	CIN->COUT (FF)	0.16165	0.48203	5.73890

Delay(ns) to SUM rising (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
FA	A->SUM (-R)	-	0.24564	0.72661	8.20872
	B->SUM (-R)	-	0.22606	0.70354	7.98273
	CIN->SUM (RR)	(A * B) + (!A * !B)	0.08919	0.46613	6.48010
	CIN->SUM (FR)	(A * !B) + (!A * B)	0.05102	0.55735	9.80895

Delay(ns) to SUM falling (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
FA	A->SUM (-F)	-	0.22320	0.58735	5.28228
	B->SUM (-F)	-	0.20383	0.56448	5.11622
	CIN->SUM (FF)	(A * B) + (!A * !B)	0.07214	0.36834	4.47672
	CIN->SUM (RF)	(A * !B) + (!A * B)	0.03154	0.40582	6.91641

Power Information

Internal switching power(pJ) to COUT rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
FA	A	17.39860	17.40860	17.76300
	B	17.39800	17.40730	17.76840
	CIN	0.00000	0.00000	0.00000
	CIN	14.35000	14.35640	14.60900

Internal switching power(pJ) to COUT falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
FA	A	5.74946	5.75350	5.87597
	B	5.75286	5.75681	5.87348
	CIN	0.00000	0.00000	0.00000
	CIN	3.75476	3.76480	4.01925

Internal switching power(pJ) to SUM rising (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
FA	A	-	0.00000	0.00000	0.00000
	A	-	26.85080	26.87130	27.54530
	B	-	0.00000	0.00000	0.00000
	B	-	26.84730	26.86870	27.54540
	CIN	(A * B) + (!A * !B)	0.00000	0.00000	0.00000
	CIN	(A * B) + (!A * !B)	27.03840	27.04640	27.28120
	CIN	(A * !B) + (!A * B)	0.00000	0.00000	0.00000
	CIN	(A * !B) + (!A * B)	3.75410	3.76520	4.02809

Internal switching power(pJ) to SUM falling (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
FA	A	-	0.00000	0.00000	0.00000
	A	-	28.06160	28.24220	29.74120
	B	-	0.00000	0.00000	0.00000
	B	-	28.05830	28.20210	29.25190
	CIN	(A * B) + (!A * !B)	0.00000	0.00000	0.00000
	CIN	(A * B) + (!A * !B)	33.67340	33.68210	33.91150
	CIN	(A * !B) + (!A * B)	0.00000	0.00000	0.00000
	CIN	(A * !B) + (!A * B)	14.35010	14.35650	14.61990

HA

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT		OUTPUT	
A	**B**	**COUT**	**SUM**
0	**0**	**0**	**0**
0	**1**	**0**	**1**
1	**0**	**0**	**1**
1	**1**	**1**	**0**

Footprint

Cell Name	Area
HA	136.80881

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)	
	A	**B**	**COUT**	**SUM**
HA	0.03316	0.03702	5.98410	2.42849

Leakage Information

Cell Name	Leakage(nW)		
	Min.	**Avg**	**Max.**
HA	0.00000	17.03700	32.46230

Delay Information

Delay(ns) to COUT rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
HA	A->COUT (RR)	0.05955	0.39116	7.60021
	B->COUT (RR)	0.05338	0.39250	7.64009

Delay(ns) to COUT falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
HA	A->COUT (FF)	0.05310	0.36995	6.17881
	B->COUT (FF)	0.04875	0.35538	6.13131

Delay(ns) to SUM rising (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
HA	A->SUM (RR)	!B	0.06372	0.43750	6.37037
	A->SUM (FR)	B	0.05832	0.57235	9.92176
	B->SUM (RR)	!A	0.08984	0.45387	6.35542
	B->SUM (FR)	A	0.04910	0.55930	9.87708

Delay(ns) to SUM falling (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
HA	A->SUM (FF)	!B	0.06429	0.35774	4.57218
	A->SUM (RF)	B	0.03468	0.37942	6.53213
	B->SUM (FF)	!A	0.07271	0.37254	4.52627
	B->SUM (RF)	A	0.02910	0.39874	6.84372

Power Information

Internal switching power(pJ) to COUT rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
HA	A	0.00000	0.00000	0.00000
	A	11.85020	11.85970	12.12750
	B	0.00000	0.00000	0.00000
	B	11.85070	11.86030	12.13490

Internal switching power(pJ) to COUT falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
HA	A	0.00000	0.00000	0.00000
	A	0.86292	0.87415	1.14188
	B	0.00000	0.00000	0.00000
	B	0.85432	0.86449	1.11019

Internal switching power(pJ) to SUM rising (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
HA	A	B	0.00000	0.00000	0.00000
	A	B	0.86293	0.87456	1.15635
	A	!B	0.00000	0.00000	0.00000
	A	!B	3.99673	4.00692	4.25922
	B	A	0.00000	0.00000	0.00000
	B	A	0.85440	0.86480	1.11966
	B	!A	0.00000	0.00000	0.00000
	B	!A	4.01184	4.01936	4.28023

Internal switching power(pJ) to SUM falling (conditional):

Cell Name	Input	When	Power(pJ)		
			first	**mid**	**last**
HA	**A**	**B**	0.00000	0.00000	0.00000
	A	**B**	11.85020	11.86070	12.14690
	A	**!B**	0.00000	0.00000	0.00000
	A	**!B**	9.43837	9.44889	9.69512
	B	**A**	0.00000	0.00000	0.00000
	B	**A**	11.85010	11.86040	12.15480
	B	**!A**	0.00000	0.00000	0.00000
	B	**!A**	9.44455	9.45315	9.67301

INVX1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT	OUTPUT
A	Y
0	1
1	0

Footprint

Cell Name	Area
INVX1	24.83200

Pin Capacitance Information

Cell Name	Pin Cap(pf)	Max Cap(pf)
	A	Y
INVX1	0.01037	5.35963

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
INVX1	0.00000	3.90439	7.80686

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
INVX1	**A->Y (FR)**	0.01514	0.45488	9.75792

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
INVX1	**A->Y (RF)**	0.01166	0.32947	7.24719

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
INVX1	A	0.00000	0.00000	0.00000
	A	0.01796	0.02309	0.06982

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
INVX1	A	0.00000	0.00000	0.00000
	A	4.60627	4.60836	4.63362

INV

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT	OUTPUT
A	Y
x	0

Footprint

Cell Name	Area
inv	4.37040

Pin Capacitance Information

Cell Name	Pin Cap(pf)	Max Cap(pf)
	A	Y
inv	0.00052	1.80000

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
inv	0.00000	0.00000	0.00000

MUX2X1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT	OUTPUT
S	**Y**
x	1

Footprint

Cell Name	Area
MUX2X1	102.35440

Pin Capacitance Information

Cell Name	Pin Cap(pf)	Max Cap(pf)
	S	**Y**
MUX2X1	0.02068	6.62826

Leakage Information

Cell Name	Leakage(nW)		
	Min.	**Avg**	**Max.**
MUX2X1	0.00000	13.86540	17.35860

NAND2X1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT		OUTPUT
A	B	Y
0	x	1
1	0	1
1	1	0

Footprint

Cell Name	Area
NAND2X1	33.44560

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	A	B	Y
NAND2X1	0.01040	0.01045	4.71037

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
NAND2X1	0.00000	3.89234	15.56570

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
NAND2X1	A->Y (FR)	0.02278	0.46412	9.47570
	B->Y (FR)	0.01817	0.45491	9.37047

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
NAND2X1	A->Y (RF)	0.02552	0.43929	9.52985
	B->Y (RF)	0.01890	0.43994	9.28271

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
NAND2X1	A	0.00000	0.00000	0.00000
	A	0.02819	0.03155	0.06589
	B	0.00000	0.00000	0.00000
	B	0.01831	0.02197	0.05221

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
NAND2X1	A	0.00000	0.00000	0.00000
	A	9.73341	9.73365	9.74491
	B	0.00000	0.00000	0.00000
	B	9.73359	9.73413	9.74944

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND2X1	(!B * Y)	0.00000	0.00000	0.00000
	(!B * Y)	-0.00178	-0.00183	-0.00181

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND2X1	(!B * Y)	0.00000	0.00000	0.00000
	(!B * Y)	0.01104	0.01060	0.01033

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND2X1	(!A * Y)	0.00000	0.00000	0.00000
	(!A * Y)	0.00110	0.00104	0.00106

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND2X1	(!A * Y)	0.00000	0.00000	0.00000
	(!A * Y)	0.01098	0.01110	0.01120

NAND3X1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT			OUTPUT
A	B	C	Y
0	x	x	1
1	0	x	1
1	1	0	1
1	1	1	0

Footprint

Cell Name	Area
NAND3X1	44.93040

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	A	B	C	Y
NAND3X1	0.01047	0.01022	0.01040	3.19774

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
NAND3X1	0.00000	2.91628	23.32410

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
NAND3X1	A->Y (FR)	0.02915	0.42890	7.83663
	B->Y (FR)	0.02602	0.41986	7.75843
	C->Y (FR)	0.02048	0.41171	7.71002

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
NAND3X1	A->Y (RF)	0.04313	0.47103	8.99082
	B->Y (RF)	0.03648	0.47105	8.97420
	C->Y (RF)	0.02782	0.47624	9.07371

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
NAND3X1	A	0.00000	0.00000	0.00000
	A	0.03781	0.04056	0.09522
	B	0.00000	0.00000	0.00000
	B	0.02882	0.03161	0.08280
	C	0.00000	0.00000	0.00000
	C	0.01969	0.02286	0.07151

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
NAND3X1	A	0.00000	0.00000	0.00000
	A	14.71980	14.71960	14.73040
	B	0.00000	0.00000	0.00000
	B	14.72070	14.72000	14.73150
	C	0.00000	0.00000	0.00000
	C	14.72040	14.71950	14.73480

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND3X1	(B * !C * Y)	0.00000	0.00000	0.00000
	(B * !C * Y)	-0.00238	-0.00240	-0.00241
	(!B * C * Y)	0.00000	0.00000	0.00000
	(!B * C * Y)	0.00071	0.00065	0.00063
	(!B * !C * Y)	0.00000	0.00000	0.00000
	(!B * !C * Y)	-0.00448	-0.00454	-0.00453

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND3X1	(B * !C * Y)	0.00000	0.00000	0.00000
	(B * !C * Y)	0.01144	0.01074	0.01055
	(!B * C * Y)	0.00000	0.00000	0.00000
	(!B * C * Y)	0.01282	0.01234	0.01220
	(!B * !C * Y)	0.00000	0.00000	0.00000
	(!B * !C * Y)	0.00992	0.00988	0.00989

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND3X1	(A * !C * Y)	0.00000	0.00000	0.00000
	(A * !C * Y)	-0.00213	-0.00215	-0.00215
	(!A * C * Y)	0.00000	0.00000	0.00000
	(!A * C * Y)	0.00425	0.00422	0.00423
	(!A * !C * Y)	0.00000	0.00000	0.00000
	(!A * !C * Y)	-0.00469	-0.00471	-0.00473

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND3X1	(A * !C * Y)	0.00000	0.00000	0.00000
	(A * !C * Y)	0.01096	0.01035	0.01026
	(!A * C * Y)	0.00000	0.00000	0.00000
	(!A * C * Y)	0.01196	0.01209	0.01211
	(!A * !C * Y)	0.00000	0.00000	0.00000
	(!A * !C * Y)	0.01097	0.01111	0.01112

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND3X1	(!B * Y)	0.00000	0.00000	0.00000
	(!B * Y)	0.00022	0.00017	0.00015
	(!A * B * Y)	0.00000	0.00000	0.00000
	(!A * B * Y)	0.00453	0.00447	0.00449

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND3X1	(!B * Y)	0.00000	0.00000	0.00000
	(!B * Y)	0.01091	0.01110	0.01113
	(!A * B * Y)	0.00000	0.00000	0.00000
	(!A * B * Y)	0.01070	0.01084	0.01080

NOR2X1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT		OUTPUT
A	B	Y
0	0	1
x	1	0
1	x	0

Footprint

Cell Name	Area
NOR2X1	33.44560

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	A	B	Y
NOR2X1	0.01065	0.01034	2.43708

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
NOR2X1	0.00000	2.00109	6.04917

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
NOR2X1	A->Y (FR)	0.03914	0.54142	9.59225
	B->Y (FR)	0.02645	0.50331	9.00793

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
NOR2X1	A->Y (RF)	0.01650	0.27809	4.81261
	B->Y (RF)	0.01392	0.26881	4.75973

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
NOR2X1	A	0.00000	0.00000	0.00000
	A	0.03223	0.03307	0.06493
	B	0.00000	0.00000	0.00000
	B	0.02677	0.03129	0.09408

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
NOR2X1	A	0.00000	0.00000	0.00000
	A	2.45426	2.45561	2.48829
	B	0.00000	0.00000	0.00000
	B	2.71950	2.71970	2.74739

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NOR2X1	(B * !Y)	0.00000	0.00000	0.00000
	(B * !Y)	0.06840	0.06676	0.06644

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NOR2X1	(B * !Y)	0.00000	0.00000	0.00000
	(B * !Y)	3.67739	3.67707	3.67697

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NOR2X1	(A * !Y)	0.00000	0.00000	0.00000
	(A * !Y)	0.01654	0.01653	0.01653

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NOR2X1	(A * !Y)	0.00000	0.00000	0.00000
	(A * !Y)	2.45788	2.45731	2.45794

NOR3X1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT			OUTPUT
A	B	C	Y
0	0	0	1
0	x	1	0
x	1	x	0
1	x	x	0

Footprint

Cell Name	Area
NOR3X1	44.93040

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	A	B	C	Y
NOR3X1	0.01073	0.01032	0.01030	1.54908

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
NOR3X1	0.00000	1.10717	5.39915

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
NOR3X1	A->Y (FR)	0.07340	0.61770	9.24835
	B->Y (FR)	0.06246	0.59747	9.15219
	C->Y (FR)	0.04110	0.55680	8.83409

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
NOR3X1	A->Y (RF)	0.01801	0.25340	3.73435
	B->Y (RF)	0.01872	0.24705	3.67714
	C->Y (RF)	0.01530	0.23612	3.63869

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
NOR3X1	A	0.00000	0.00000	0.00000
	A	0.04601	0.04602	0.07663
	B	0.00000	0.00000	0.00000
	B	0.04074	0.04107	0.07544
	C	0.00000	0.00000	0.00000
	C	0.03571	0.03869	0.10021

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
NOR3X1	A	0.00000	0.00000	0.00000
	A	2.50853	2.50977	2.56233
	B	0.00000	0.00000	0.00000
	B	1.81912	1.81949	1.85487
	C	0.00000	0.00000	0.00000
	C	1.87836	1.87835	1.90828

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NOR3X1	(B * C * !Y)	0.00000	0.00000	0.00000
	(B * C * !Y)	0.01891	0.01811	0.01794
	(B * !C * !Y)	0.00000	0.00000	0.00000
	(B * !C * !Y)	0.09053	0.08899	0.08868
	(!B * C * !Y)	0.00000	0.00000	0.00000
	(!B * C * !Y)	0.06634	0.06450	0.06414

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NOR3X1	(B * C * !Y)	0.00000	0.00000	0.00000
	(B * C * !Y)	0.06877	0.06883	0.06887
	(B * !C * !Y)	0.00000	0.00000	0.00000
	(B * !C * !Y)	1.79557	1.79562	1.79560
	(!B * C * !Y)	0.00000	0.00000	0.00000
	(!B * C * !Y)	3.07937	3.07951	3.07974

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NOR3X1	(A * C * !Y)	0.00000	0.00000	0.00000
	(A * C * !Y)	0.02230	0.02229	0.02232
	(A * !C * !Y)	0.00000	0.00000	0.00000
	(A * !C * !Y)	0.04625	0.04624	0.04625
	(!A * C * !Y)	0.00000	0.00000	0.00000
	(!A * C * !Y)	0.04451	0.04260	0.04222

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NOR3X1	(A * C * !Y)	0.00000	0.00000	0.00000
	(A * C * !Y)	0.07854	0.07857	0.07861
	(A * !C * !Y)	0.00000	0.00000	0.00000
	(A * !C * !Y)	2.50812	2.50789	2.50843
	(!A * C * !Y)	0.00000	0.00000	0.00000
	(!A * C * !Y)	3.11735	3.11706	3.11679

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NOR3X1	(B * !Y)	0.00000	0.00000	0.00000
	(B * !Y)	0.00814	0.00813	0.00813
	(A * !B * !Y)	0.00000	0.00000	0.00000
	(A * !B * !Y)	0.01196	0.01196	0.01196

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NOR3X1	(B * !Y)	0.00000	0.00000	0.00000
	(B * !Y)	0.11681	0.11680	0.11684
	(A * !B * !Y)	0.00000	0.00000	0.00000
	(A * !B * !Y)	2.50974	2.50930	2.50983

OR2X1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT		OUTPUT
A	B	Y
0	0	0
x	1	1
1	x	1

Footprint

Cell Name	Area
OR2X1	50.67280

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	A	B	Y
OR2X1	0.01061	0.01049	5.52141

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
OR2X1	0.00000	3.08380	5.40294

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
OR2X1	A->Y (RR)	0.04516	0.33856	6.37444
	B->Y (RR)	0.04091	0.34007	6.63750

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
OR2X1	A->Y (FF)	0.08209	0.39259	6.17315
	B->Y (FF)	0.06861	0.38356	6.17202

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
OR2X1	A	0.00000	0.00000	0.00000
	A	1.85445	1.86697	2.17494
	B	0.00000	0.00000	0.00000
	B	2.93773	2.94701	3.19267

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
OR2X1	A	0.00000	0.00000	0.00000
	A	1.25377	1.25771	1.46869
	B	0.00000	0.00000	0.00000
	B	2.95760	2.96721	3.23461

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR2X1	(B * Y)	0.00000	0.00000	0.00000
	(B * Y)	0.04920	0.04745	0.04723

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR2X1	(B * Y)	0.00000	0.00000	0.00000
	(B * Y)	3.12388	3.12267	3.12347

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR2X1	(A * Y)	0.00000	0.00000	0.00000
	(A * Y)	0.01965	0.01965	0.01965

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR2X1	(A * Y)	0.00000	0.00000	0.00000
	(A * Y)	1.84543	1.84523	1.84579

OR3X1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT			OUTPUT
A	B	C	Y
0	0	0	0
0	x	1	1
x	1	x	1
1	x	x	1

Footprint

Cell Name	Area
OR3X1	62.15760

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	A	B	C	Y
OR3X1	0.01062	0.01042	0.01040	5.44756

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
OR3X1	0.00000	1.61261	5.01932

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
OR3X1	A->Y (RR)	0.04920	0.34877	6.20705
	B->Y (RR)	0.04815	0.35052	6.18156
	C->Y (RR)	0.04287	0.34584	6.47297

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
OR3X1	A->Y (FF)	0.13204	0.45152	6.28705
	B->Y (FF)	0.12048	0.45446	6.32742
	C->Y (FF)	0.09735	0.44093	6.33674

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
OR3X1	A	0.00000	0.00000	0.00000
	A	1.89557	1.90618	2.20265
	B	0.00000	0.00000	0.00000
	B	1.61657	1.62328	1.83786
	C	0.00000	0.00000	0.00000
	C	2.55296	2.55947	2.74913

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
OR3X1	A	0.00000	0.00000	0.00000
	A	0.62599	0.62800	0.79737
	B	0.00000	0.00000	0.00000
	B	1.31342	1.31585	1.48283
	C	0.00000	0.00000	0.00000
	C	2.58849	2.59516	2.80552

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR3X1	(B * C * Y)	0.00000	0.00000	0.00000
	(B * C * Y)	0.01832	0.01749	0.01736
	(B * !C * Y)	0.00000	0.00000	0.00000
	(B * !C * Y)	0.06606	0.06447	0.06427
	(!B * C * Y)	0.00000	0.00000	0.00000
	(!B * C * Y)	0.04944	0.04759	0.04728

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR3X1	(B * C * Y)	0.00000	0.00000	0.00000
	(B * C * Y)	0.06098	0.06104	0.06107
	(B * !C * Y)	0.00000	0.00000	0.00000
	(B * !C * Y)	1.53497	1.53511	1.53509
	(!B * C * Y)	0.00000	0.00000	0.00000
	(!B * C * Y)	2.67803	2.67751	2.67775

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR3X1	(A * C * Y)	0.00000	0.00000	0.00000
	(A * C * Y)	0.02084	0.02085	0.02088
	(A * !C * Y)	0.00000	0.00000	0.00000
	(A * !C * Y)	0.03884	0.03883	0.03884
	(!A * C * Y)	0.00000	0.00000	0.00000
	(!A * C * Y)	0.03922	0.03717	0.03695

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR3X1	(A * C * Y)	0.00000	0.00000	0.00000
	(A * C * Y)	0.06442	0.06445	0.06447
	(A * !C * Y)	0.00000	0.00000	0.00000
	(A * !C * Y)	1.87981	1.87978	1.88007
	(!A * C * Y)	0.00000	0.00000	0.00000
	(!A * C * Y)	2.75138	2.75076	2.75130

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR3X1	(B * Y)	0.00000	0.00000	0.00000
	(B * Y)	0.01249	0.01248	0.01247
	(A * !B * Y)	0.00000	0.00000	0.00000
	(A * !B * Y)	0.01714	0.01714	0.01712

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR3X1	(B * Y)	0.00000	0.00000	0.00000
	(B * Y)	0.08812	0.08813	0.08818
	(A * !B * Y)	0.00000	0.00000	0.00000
	(A * !B * Y)	1.88327	1.88308	1.88367

TIEHI

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Footprint

Cell Name	Area
TIEHI	24.83200

Pin Capacitance Information

Cell Name	Max Cap(pf)
	Y
TIEHI	11.48078

Leakage Information

Cell Name	Leakage(nW)		
	Min.	**Avg**	**Max.**
TIEHI	0.00000	0.00000	0.00000

TIELO

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Footprint

Cell Name	Area
TIELO	24.83200

Pin Capacitance Information

Cell Name	Max Cap(pf)
	YN
TIELO	21.89425

Leakage Information

Cell Name	Leakage(nW)		
	Min.	**Avg**	**Max.**
TIELO	0.00000	0.00000	0.00000

TMRDFFQNX1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT		OUTPUT
D	CLK	QN
0	R	1
1	R	0
x	x	IQN

Footprint

Cell Name	Area
TMRDFFQNX1	584.71600

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	D	CLK	QN
TMRDFFQNX1	0.03877	0.07288	3.18283

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
TMRDFFQNX1	0.00000	80.34340	116.88600

Delay Information

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
TMRDFFQNX1	CLK->QN (RR)	0.34857	0.78595	7.68437

Delay(ns) to QN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
TMRDFFQNX1	CLK->QN (RF)	0.19831	0.41566	2.16562

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFQNX1	hold	CLK (R)	0.04756	0.07077	0.63804
	setup	CLK (R)	0.10773	0.14408	0.75756

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFQNX1	hold	CLK (R)	-0.03491	-0.07666	-0.72062
	setup	CLK (R)	0.05518	0.10001	1.37315

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFQNX1	min_pulse_width	CLK ()	D	0.14511	0.47852	16.50020
	min_pulse_width	CLK ()	!D	0.14874	0.47852	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFQNX1	min_pulse_width	CLK ()	D	0.19585	0.47852	16.50020
	min_pulse_width	CLK ()	!D	0.09075	0.47852	16.50020

Power Information

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
TMRDFFQNX1	CLK	0.00000	0.00000	0.00000
	CLK	74.71390	74.71870	75.38450

Internal switching power(pJ) to QN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
TMRDFFQNX1	CLK	0.00000	0.00000	0.00000
	CLK	63.43570	63.43760	64.05990

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFQNX1	(CLK * QN)	0.00000	0.00000	0.00000
	(CLK * QN)	70.26790	70.27090	70.27230
	(CLK * !QN)	0.00000	0.00000	0.00000
	(CLK * !QN)	74.51070	74.51570	74.93980
	!CLK	0.00000	0.00000	0.00000
	!CLK	31.62340	31.62870	32.24810

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFQNX1	(CLK * QN)	0.00000	0.00000	0.00000
	(CLK * QN)	72.12910	72.12860	72.13090
	(CLK * !QN)	0.00000	0.00000	0.00000
	(CLK * !QN)	50.49810	50.50210	50.94280
	!CLK	0.00000	0.00000	0.00000
	!CLK	31.67120	31.68650	32.32680

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFQNX1	(D * !QN)	0.00000	0.00000	0.00000
	(D * !QN)	63.65260	63.65300	64.26300
	(!D * QN)	0.00000	0.00000	0.00000
	(!D * QN)	74.29170	74.29320	74.88100

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFQNX1	(D * QN)	0.00000	0.00000	0.00000
	(D * QN)	29.44040	29.44380	30.06230
	(D * !QN)	0.00000	0.00000	0.00000
	(D * !QN)	36.53270	36.53670	37.17820
	(!D * QN)	0.00000	0.00000	0.00000
	(!D * QN)	29.76280	29.77280	30.40080
	(!D * !QN)	0.00000	0.00000	0.00000
	(!D * !QN)	35.01800	35.02790	35.66230

TMRDFFQX1

sky130_rhbd_tt_1P8_25C.ccs Cell Library:
Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT		OUTPUT
D	CLK	Q
0	R	0
1	R	1
x	x	IQ

Footprint

Cell Name	Area
TMRDFFQX1	601.94318

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	D	CLK	Q
TMRDFFQX1	0.03876	0.07295	5.23553

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
TMRDFFQX1	0.00000	83.30790	123.69400

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
TMRDFFQX1	**CLK->Q (RR)**	0.23749	0.61296	7.11888

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
TMRDFFQX1	**CLK->Q (RF)**	0.39889	0.71022	5.47312

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFQX1	hold	CLK (R)	0.04559	0.06928	0.59673
	setup	CLK (R)	0.10525	0.13502	0.72764

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFQX1	hold	CLK (R)	-0.03836	-0.07514	-0.71034
	setup	CLK (R)	0.05305	0.10001	1.35865

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFQX1	min_pulse_width	CLK ()	D	0.14874	0.47852	16.50020
	min_pulse_width	CLK ()	!D	0.15236	0.47852	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFQX1	min_pulse_width	CLK ()	D	0.19222	0.47852	16.50020
	min_pulse_width	CLK ()	!D	0.09075	0.47852	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
TMRDFFQX1	CLK	0.00000	0.00000	0.00000
	CLK	62.84680	62.84910	63.48220

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
TMRDFFQX1	CLK	0.00000	0.00000	0.00000
	CLK	78.99330	79.00130	79.61620

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFQX1	(CLK * Q)	0.00000	0.00000	0.00000
	(CLK * Q)	73.77440	73.77950	74.20280
	(CLK * !Q)	0.00000	0.00000	0.00000
	(CLK * !Q)	74.45620	74.46340	74.46520
	!CLK	0.00000	0.00000	0.00000
	!CLK	35.23910	35.24450	35.86490

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFQX1	(CLK * Q)	0.00000	0.00000	0.00000
	(CLK * Q)	50.03470	50.04300	50.47930
	(CLK * !Q)	0.00000	0.00000	0.00000
	(CLK * !Q)	76.32480	76.32790	76.33020
	!CLK	0.00000	0.00000	0.00000
	!CLK	35.29070	35.30340	35.94400

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFQX1	(D * Q)	0.00000	0.00000	0.00000
	(D * Q)	63.01590	63.01670	63.62510
	(!D * !Q)	0.00000	0.00000	0.00000
	(!D * !Q)	78.59380	78.59230	79.18340

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFQX1	(D * Q)	0.00000	0.00000	0.00000
	(D * Q)	36.22660	36.22950	36.87230
	(D * !Q)	0.00000	0.00000	0.00000
	(D * !Q)	32.84850	32.85350	33.47340
	(!D * Q)	0.00000	0.00000	0.00000
	(!D * Q)	34.64970	34.65830	35.29350
	(!D * !Q)	0.00000	0.00000	0.00000
	(!D * !Q)	33.21240	33.22100	33.84840

TMRDFFRNQNX1

sky130_rhbd_tt_1P8_25C.ccs
Cell Library: Process , Voltage
1.80, Temp 25.00

Truth Table

INPUT			OUTPUT
D	RN	CLK	QN
x	x	x	-

Footprint

Cell Name	Area
TMRDFFRNQNX1	688.07922

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	D	RN	CLK	QN
TMRDFFRNQNX1	0.03984	0.08896	0.06778	1.92652

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
TMRDFFRNQNX1	0.00000	599527.00000	944127.00000

Delay Information
Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
TMRDFFRNQNX1	**CLK->QN (FR)**	0.51598	1.24450	13.40230

Constraint Information
Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFRNQNX1	**min_pulse_width**	**CLK ()**	**(!D * RN)**	0.04792	0.54696	16.50020

Power Information

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
TMRDFFRNQNX1	CLK	0.00000	0.00000	0.00000
	CLK	484075.00000	484073.00000	484093.00000

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFRNQNX1	(CLK * RN * !QN)	0.00000	0.00000	0.00000
	(CLK * RN * !QN)	403916.00000	403915.00000	403913.00000
	(!CLK * RN * QN)	0.00000	0.00000	0.00000
	(!CLK * RN * QN)	430238.00000	430238.00000	430239.00000

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFRNQNX1	(!CLK * RN * QN)	0.00000	0.00000	0.00000
	(!CLK * RN * QN)	484299.00000	484299.00000	484303.00000

TMRDFFRNQX1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT			OUTPUT
D	**RN**	**CLK**	**Q**
x	x	x	-

Footprint

Cell Name	Area
TMRDFFRNQX1	705.30640

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	D	**RN**	**CLK**	**Q**
TMRDFFRNQX1	0.03983	0.08899	0.06783	8.24007

Leakage Information

Cell Name	Leakage(nW)		
	Min.	**Avg**	**Max.**
TMRDFFRNQX1	0.00000	599971.00000	944690.00000

Delay Information
Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	**Mid**	**Last**
TMRDFFRNQX1	**CLK->Q (FF)**	0.55848	1.16649	13.63140

Constraint Information

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFRNQX1	**min_pulse_width**	**CLK ()**	**(!D * RN)**	0.06168	0.53812	16.50020

Power Information

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
TMRDFFRNQX1	CLK	-0.00227	-0.41417	-13.32100
	CLK	484356.00000	484353.00000	484375.00000

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFRNQX1	(CLK * RN * Q)	0.00000	0.00000	0.00000
	(CLK * RN * Q)	404174.00000	404172.00000	404171.00000
	(!CLK * RN * !Q)	0.00000	0.00000	0.00000
	(!CLK * RN * !Q)	430767.00000	430767.00000	430768.00000

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFRNQX1	(!CLK * RN * !Q)	0.00000	0.00000	0.00000
	(!CLK * RN * !Q)	484601.00000	484601.00000	484605.00000

TMRDFFSNQNX1

sky130_rhbd_tt_1P8_25C.ccs
Cell Library: Process , Voltage
1.80, Temp 25.00

Truth Table

INPUT			OUTPUT
D	SN	CLK	QN
0	1	R	0
1	1	R	1
x	0	x	1
x	1	x	IQN

Footprint

Cell Name	Area
TMRDFFSNQNX1	653.62482

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	D	SN	CLK	QN
TMRDFFSNQNX1	0.04043	0.06714	0.07093	3.08854

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
TMRDFFSNQNX1	0.00000	85.23720	151.46700

Delay Information

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
TMRDFFSNQNX1	CLK->QN (RR)	0.31801	0.77133	7.50363
	SN->QN (FR)	0.39728	0.87141	8.19751

Delay(ns) to QN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
TMRDFFSNQNX1	CLK->QN (RF)	0.22845	0.42359	2.14794

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFSNQNX1	hold	CLK (R)	0.01334	0.01557	0.69268
	setup	CLK (R)	0.12380	0.15124	0.92660

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFSNQNX1	hold	CLK (R)	-0.07786	-0.13252	-1.05589
	setup	CLK (R)	0.10629	0.17055	1.59972

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNQNX1	hold	CLK (R)	SN	0.01334	0.01557	0.69268
	setup	CLK (R)	SN	0.12380	0.15124	0.92660

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNQNX1	hold	CLK (R)	SN	-0.07786	-0.13252	-1.05589
	setup	CLK (R)	SN	0.10629	0.17055	1.59972

Constraints(ns) for SN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFSNQNX1	recovery	CLK (R)	0.03491	0.03477	0.28436
	removal	CLK (R)	-0.01438	-0.01250	-0.10434

Constraints(ns) for SN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNQNX1	recovery	CLK (R)	!D	0.03491	0.03477	0.28436
	removal	CLK (R)	!D	-0.01438	-0.01250	-0.10434

Constraints(ns) for SN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNQNX1	min_pulse_width	SN ()	(CLK * D)	0.07626	0.47852	16.50020
	min_pulse_width	SN ()	(CLK * !D)	0.07626	0.47852	16.50020
	min_pulse_width	SN ()	(!CLK * D)	0.09075	0.47852	16.50020
	min_pulse_width	SN ()	(!CLK * !D)	0.09075	0.47852	16.50020

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNQNX1	min_pulse_width	CLK ()	(D * SN)	0.12699	0.47852	16.50020
	min_pulse_width	CLK ()	(!D * SN)	0.18135	0.47852	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNQNX1	min_pulse_width	CLK ()	(D * SN)	0.20309	0.47852	16.50020
	min_pulse_width	CLK ()	(!D * SN)	0.11249	0.47852	16.50020

Power Information

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
TMRDFFSNQNX1	CLK	0.00000	0.00000	0.00000
	CLK	56.55320	56.56630	57.25990
	SN	54.93480	54.96780	55.96830

Internal switching power(pJ) to QN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
TMRDFFSNQNX1	CLK	0.00000	0.00000	0.00000
	CLK	105.54200	105.54800	106.14700

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQNX1	(CLK * SN * !QN)	0.00000	0.00000	0.00000
	(CLK * SN * !QN)	101.11300	101.11400	101.11400
	(CLK * QN) + (!CLK * !SN * QN)	0.00000	0.00000	0.00000
	(CLK * QN) + (!CLK * !SN * QN)	69.04350	69.04490	69.43770
	(!CLK * SN)	0.00000	0.00000	0.00000
	(!CLK * SN)	32.82000	32.82420	33.41570

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQNX1	(CLK * SN * !QN)	0.00000	0.00000	0.00000
	(CLK * SN * !QN)	102.95600	102.95500	102.95400
	(CLK * QN) + (!CLK * !SN * QN)	0.00000	0.00000	0.00000
	(CLK * QN) + (!CLK * !SN * QN)	47.85100	47.85450	48.26920
	(!CLK * SN)	0.00000	0.00000	0.00000
	(!CLK * SN)	46.31230	46.33050	46.95040

Passive power(pJ) for SN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQNX1	(CLK * QN) + (!CLK * D * QN)	0.00000	0.00000	0.00000
	(CLK * QN) + (!CLK * D * QN)	51.92810	51.93540	51.94780
	(!CLK * !D * QN)	0.00000	0.00000	0.00000
	(!CLK * !D * QN)	39.59660	39.59910	40.15390

Passive power(pJ) for SN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQNX1	(CLK * QN) + (!CLK * D * QN)	0.00000	0.00000	0.00000
	(CLK * QN) + (!CLK * D * QN)	55.79060	55.79440	56.10280
	(!CLK * !D * QN)	0.00000	0.00000	0.00000
	(!CLK * !D * QN)	10.94280	10.96440	11.75650

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQNX1	(D * QN)	0.00000	0.00000	0.00000
	(D * QN)	56.42850	56.43560	57.08060
	(!D * SN * !QN)	0.00000	0.00000	0.00000
	(!D * SN * !QN)	105.21200	105.21300	105.79400
	(!D * !SN * QN)	0.00000	0.00000	0.00000
	(!D * !SN * QN)	29.16560	29.18870	29.95270

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQNX1	(D * SN * !QN)	0.00000	0.00000	0.00000
	(D * SN * !QN)	40.71970	40.71860	41.40760
	(D * QN)	0.00000	0.00000	0.00000
	(D * QN)	30.74660	30.74360	31.46490
	(!D * SN * QN)	0.00000	0.00000	0.00000
	(!D * SN * QN)	45.34650	45.35930	46.03420
	(!D * SN * !QN)	0.00000	0.00000	0.00000
	(!D * SN * !QN)	55.26760	55.28080	55.92530
	(!D * !SN * QN)	0.00000	0.00000	0.00000
	(!D * !SN * QN)	10.11050	10.13150	10.99680

TMRDFFSNQX1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT			OUTPUT
D	SN	CLK	Q
0	1	R	0
1	1	R	1
x	0	x	1
x	1	x	IQ

Footprint

Cell Name	Area
TMRDFFSNQX1	670.85199

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	D	SN	CLK	Q
TMRDFFSNQX1	0.04037	0.06959	0.07087	5.19233

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
TMRDFFSNQX1	0.00000	85.20620	149.46600

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
TMRDFFSNQX1	CLK->Q (RR)	0.23608	0.60575	7.04065
	SN->Q (FR)	0.14643	0.57489	7.89148

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
TMRDFFSNQX1	CLK->Q (RF)	0.44617	0.75923	5.53210

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFSNQX1	hold	CLK (R)	0.01132	0.01000	0.49133
	setup	CLK (R)	0.13022	0.16551	0.97657

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFSNQX1	hold	CLK (R)	-0.07275	-0.13402	-1.10975
	setup	CLK (R)	0.09839	0.16002	1.46160

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNQX1	hold	CLK (R)	SN	0.01132	0.01000	0.49133
	setup	CLK (R)	SN	0.13022	0.16551	0.97657

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNQX1	hold	CLK (R)	SN	-0.07275	-0.13402	-1.10975
	setup	CLK (R)	SN	0.09839	0.16002	1.46160

Constraints(ns) for SN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFSNQX1	recovery	CLK (R)	0.02662	0.02626	3.84614
	removal	CLK (R)	-0.01685	-0.01250	-0.10876

Constraints(ns) for SN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNQX1	recovery	CLK (R)	!D	0.02662	0.02626	3.84614
	removal	CLK (R)	!D	-0.01685	-0.01250	-0.10876

Constraints(ns) for SN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNQX1	min_pulse_width	SN ()	(CLK * D)	0.09438	0.47852	16.50020
	min_pulse_width	SN ()	(CLK * !D)	0.09438	0.47852	16.50020
	min_pulse_width	SN ()	(!CLK * D)	0.09800	0.47852	16.50020
	min_pulse_width	SN ()	(!CLK * !D)	0.09800	0.47852	16.50020

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNQX1	min_pulse_width	CLK ()	(D * SN)	0.14511	0.47852	16.50020
	min_pulse_width	CLK ()	(!D * SN)	0.15961	0.47852	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNQX1	min_pulse_width	CLK ()	(D * SN)	0.20672	0.47852	16.50020
	min_pulse_width	CLK ()	(!D * SN)	0.10162	0.47852	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
TMRDFFSNQX1	CLK	0.00000	0.00000	0.00000
	CLK	58.91030	58.91450	59.56510
	SN	57.80780	57.82400	58.46300

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
TMRDFFSNQX1	CLK	0.00000	0.00000	0.00000
	CLK	105.54500	105.55000	106.18300

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQX1	(CLK * SN * !Q)	0.00000	0.00000	0.00000
	(CLK * SN * !Q)	100.05300	100.06100	100.06200
	(CLK * Q) + (!CLK * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * Q) + (!CLK * !SN * Q)	79.38780	79.39000	79.78540
	(!CLK * SN)	0.00000	0.00000	0.00000
	(!CLK * SN)	40.17140	40.17550	40.76510

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQX1	(CLK * SN * !Q)	0.00000	0.00000	0.00000
	(CLK * SN * !Q)	101.96500	101.96800	101.97000
	(CLK * Q) + (!CLK * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * Q) + (!CLK * !SN * Q)	55.90140	55.90940	56.31600
	(!CLK * SN)	0.00000	0.00000	0.00000
	(!CLK * SN)	54.53820	54.55750	55.17870

Passive power(pJ) for SN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQX1	(CLK * Q) + (!CLK * D * Q)	0.00000	0.00000	0.00000
	(CLK * Q) + (!CLK * D * Q)	55.42040	55.43150	55.43210
	(!CLK * !D * Q)	0.00000	0.00000	0.00000
	(!CLK * !D * Q)	44.35800	44.35760	44.92910

Passive power(pJ) for SN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQX1	(CLK * Q) + (!CLK * D * Q)	0.00000	0.00000	0.00000
	(CLK * Q) + (!CLK * D * Q)	58.77330	58.78150	58.78380
	(!CLK * !D * Q)	0.00000	0.00000	0.00000
	(!CLK * !D * Q)	12.09570	12.11300	12.72160

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQX1	(D * Q)	0.00000	0.00000	0.00000
	(D * Q)	59.45960	59.46310	60.10080
	(!D * SN * !Q)	0.00000	0.00000	0.00000
	(!D * SN * !Q)	104.88000	104.87800	105.47700
	(!D * !SN * Q)	0.00000	0.00000	0.00000
	(!D * !SN * Q)	33.17050	33.19390	33.97990

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQX1	(D * SN * !Q)	0.00000	0.00000	0.00000
	(D * SN * !Q)	40.86400	40.85690	41.56090
	(D * Q)	0.00000	0.00000	0.00000
	(D * Q)	37.77410	37.77600	38.48180
	(!D * SN * Q)	0.00000	0.00000	0.00000
	(!D * SN * Q)	54.54320	54.55120	55.23380
	(!D * SN * !Q)	0.00000	0.00000	0.00000
	(!D * SN * !Q)	53.49960	53.50010	54.12880
	(!D * !SN * Q)	0.00000	0.00000	0.00000
	(!D * !SN * Q)	11.60070	11.62170	12.47400

TMRDFFSNRNQNX1

sky130_rhbd_tt_1P8_25C.ccs
Cell Library: Process ,
Voltage 1.80, Temp 25.00

Truth Table

INPUT				OUTPUT
D	RN	SN	CLK	Q
0	1	1	R	0
1	1	1	R	1
x	0	x	x	0
x	1	0	x	1
x	1	1	x	IQ

Footprint

Cell Name	Area
TMRDFFSNRNQNX1	756.98798

Pin Capacitance Information

Cell Name	Pin Cap(pf)				Max Cap(pf)
	D	RN	SN	CLK	Q
TMRDFFSNRNQNX1	0.04195	0.10404	0.07578	0.07531	3.09222

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
TMRDFFSNRNQNX1	0.00000	97.39300	168.17500

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
TMRDFFSNRNQNX1	CLK->Q (RR)	0.41513	0.87180	7.70007
	RN->Q (RR)	0.41851	0.90804	8.34347
	SN->Q (FR)	0.28776	0.77561	8.28851

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
TMRDFFSNRNQNX1	CLK->Q (RF)	0.23447	0.43710	2.20357
	RN->Q (FF)	0.11407	0.39078	3.02427

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFSNRNQNX1	hold	CLK (R)	-0.01473	-0.01162	1.95543
	setup	CLK (R)	0.15888	0.17213	0.63283

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFSNRNQNX1	hold	CLK (R)	-0.07236	-0.12802	-1.00451
	setup	CLK (R)	0.11340	0.17730	2.09475

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNRNQNX1	hold	CLK (R)	(RN * SN)	-0.01473	-0.01162	1.95543
	setup	CLK (R)	(RN * SN)	0.15888	0.17213	0.63283

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNRNQNX1	hold	CLK (R)	(RN * SN)	-0.07236	-0.12802	-1.00451
	setup	CLK (R)	(RN * SN)	0.11340	0.17730	2.09475

Constraints(ns) for RN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFSNRNQNX1	recovery	CLK (R)	0.14919	0.17578	6.74907
	removal	CLK (R)	-0.01918	-0.02500	-0.06836
	hold	SN (R)	0.00881	-0.01000	-0.11322
	setup	SN (R)	0.05993	0.10001	0.93397

Constraints(ns) for RN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNRNQNX1	recovery	CLK (R)	(D * SN)	0.14919	0.17578	6.74907
	removal	CLK (R)	(D * SN)	-0.01918	-0.02500	-0.06836
	hold	SN (R)	(CLK * D)	0.00783	-0.01000	-0.11322
	hold	SN (R)	(CLK * !D)	0.00881	-0.01250	-0.11727
	hold	SN (R)	(!CLK * D)	-0.04516	-0.08001	-0.23327
	hold	SN (R)	(!CLK * !D)	-0.04669	-0.08001	-0.23684
	setup	SN (R)	(CLK * D)	0.00181	0.01750	0.28551
	setup	SN (R)	(CLK * !D)	-0.00066	0.01750	0.30870
	setup	SN (R)	(!CLK * D)	0.05993	0.10001	0.74134
	setup	SN (R)	(!CLK * !D)	0.05684	0.10001	0.93397

Constraints(ns) for RN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNRNQNX1	min_pulse_width	RN ()	(CLK * D * SN)	0.11249	0.47852	16.50020
	min_pulse_width	RN ()	(CLK * !D * SN)	0.10162	0.47852	16.50020
	min_pulse_width	RN ()	(!CLK * D * SN)	0.10525	0.47852	16.50020
	min_pulse_width	RN ()	(!CLK * !D * SN)	0.10525	0.47852	16.50020

Constraints(ns) for SN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFSNRNQNX1	recovery	CLK (R)	0.02619	0.01909	0.23950
	removal	CLK (R)	0.00266	0.01000	0.01752
	hold	RN (R)	0.06609	0.11752	0.50681
	setup	RN (R)	0.02340	0.00797	0.64833

Constraints(ns) for SN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNRNQNX1	recovery	CLK (R)	(!D * RN)	0.02619	0.01909	0.23950
	removal	CLK (R)	(!D * RN)	0.00266	0.01000	0.01752
	hold	RN (R)	CLK	0.02519	0.06751	0.36038
	hold	RN (R)	(!CLK * D)	0.06609	0.11502	0.48246
	hold	RN (R)	(!CLK * !D)	0.06606	0.11752	0.50681
	setup	RN (R)	CLK	0.02340	0.00797	0.57565
	setup	RN (R)	(!CLK * D)	-0.03422	-0.06820	0.64833
	setup	RN (R)	(!CLK * !D)	-0.03455	-0.07277	0.08215

Constraints(ns) for SN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNRNQNX1	min_pulse_width	SN ()	(CLK * D * RN)	0.09438	0.47852	16.50020
	min_pulse_width	SN ()	(CLK * !D * RN)	0.09438	0.47852	16.50020
	min_pulse_width	SN ()	(!CLK * D * RN)	0.11249	0.47852	16.50020
	min_pulse_width	SN ()	(!CLK * !D * RN)	0.11249	0.47852	16.50020

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNRNQNX1	min_pulse_width	CLK ()	(D * RN * SN)	0.18498	0.47852	16.50020
	min_pulse_width	CLK ()	(!D * RN * SN)	0.19222	0.47852	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNRNQNX1	min_pulse_width	CLK ()	(D * RN * SN)	0.22484	0.47852	16.50020
	min_pulse_width	CLK ()	(!D * RN * SN)	0.12337	0.47852	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
TMRDFFSNRNQNX1	CLK	0.00000	0.00000	0.00000
	CLK	94.34000	94.35150	94.96630
	RN	95.16580	95.17820	96.31050
	SN	94.17060	94.20270	95.18000

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
TMRDFFSNRNQNX1	CLK	0.00000	0.00000	0.00000
	CLK	97.44780	97.45100	98.02660
	RN	-0.00228	-0.23051	-5.00940
	RN	98.98950	99.02090	100.04400

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNRNQNX1	(CLK * RN * SN * !Q) + (CLK * !RN * !Q) + (!CLK * !RN * !Q)	0.00000	0.00000	0.00000
	(CLK * RN * SN * !Q) + (CLK * !RN * !Q) + (!CLK * !RN * !Q)	99.64850	99.64790	99.83210
	(CLK * RN * Q) + (!CLK * RN * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * RN * Q) + (!CLK * RN * !SN * Q)	93.55910	93.55970	94.12880
	(!CLK * RN * SN)	0.00000	0.00000	0.00000
	(!CLK * RN * SN)	50.18120	50.18120	50.75480

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNRNQNX1	(CLK * RN * SN * !Q) + (CLK * !RN * !Q) + (!CLK * !RN * !Q)	0.00000	0.00000	0.00000
	(CLK * RN * SN * !Q) + (CLK * !RN * !Q) + (!CLK * !RN * !Q)	95.53420	95.53080	95.53100
	(CLK * RN * Q) + (!CLK * RN * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * RN * Q) + (!CLK * RN * !SN * Q)	59.88710	59.89190	60.30400
	(!CLK * RN * SN)	0.00000	0.00000	0.00000
	(!CLK * RN * SN)	50.24980	50.26470	50.90050

Passive power(pJ) for RN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNRNQNX1	(CLK * SN * !Q) + (!CLK * !D * SN * !Q)	0.00000	0.00000	0.00000
	(CLK * SN * !Q) + (!CLK * !D * SN * !Q)	91.68680	91.68210	91.68280
	(!CLK * D * SN * !Q)	0.00000	0.00000	0.00000
	(!CLK * D * SN * !Q)	51.93040	51.93280	52.48430

Passive power(pJ) for RN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNRNQNX1	(CLK * SN * !Q) + (!CLK * !D * SN * !Q)	0.00000	0.00000	0.00000
	(CLK * SN * !Q) + (!CLK * !D * SN * !Q)	96.98140	96.98150	96.98170
	(!CLK * D * SN * !Q)	0.00000	0.00000	0.00000
	(!CLK * D * SN * !Q)	55.22790	55.23730	55.81400

Passive power(pJ) for SN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNRNQNX1	(CLK * RN * Q) + (!CLK * D * RN * Q)	0.00000	0.00000	0.00000
	(CLK * RN * Q) + (!CLK * D * RN * Q)	88.02430	88.05220	88.05560
	(!RN * !Q)	0.00000	0.00000	0.00000
	(!RN * !Q)	98.45350	98.46410	99.62060
	(!CLK * !D * RN * Q)	0.00000	0.00000	0.00000
	(!CLK * !D * RN * Q)	49.42240	49.42220	49.98630

Passive power(pJ) for SN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNRNQNX1	(CLK * RN * Q) + (!CLK * D * RN * Q)	0.00000	0.00000	0.00000
	(CLK * RN * Q) + (!CLK * D * RN * Q)	92.91090	92.91740	93.22120
	(!RN * !Q)	0.00000	0.00000	0.00000
	(!RN * !Q)	16.48370	16.51980	17.77730
	(!CLK * !D * RN * Q)	0.00000	0.00000	0.00000
	(!CLK * !D * RN * Q)	18.08110	18.10410	18.89170

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNRNQNX1	(D * RN * Q)	0.00000	0.00000	0.00000
	(D * RN * Q)	93.97090	93.97030	94.54690
	(D * !RN * !Q) + (!D * RN * SN * !Q) + (!D * !RN * !Q)	0.00000	0.00000	0.00000
	(D * !RN * !Q) + (!D * RN * SN * !Q) + (!D * !RN * !Q)	97.55390	97.55650	98.11190
	(!D * RN * !SN * Q)	0.00000	0.00000	0.00000
	(!D * RN * !SN * Q)	51.19880	51.21080	51.94930

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNRNQNX1	(D * RN * SN * !Q)	0.00000	0.00000	0.00000
	(D * RN * SN * !Q)	54.10790	54.11840	54.74880
	(D * RN * Q)	0.00000	0.00000	0.00000
	(D * RN * Q)	46.36330	46.36390	47.02300
	(D * !RN * !Q) + (!D * RN * SN * !Q) + (!D * !RN * !Q)	0.00000	0.00000	0.00000
	(D * !RN * !Q) + (!D * RN * SN * !Q) + (!D * !RN * !Q)	55.15810	55.17230	55.81390
	(!D * RN * SN * Q)	0.00000	0.00000	0.00000
	(!D * RN * SN * Q)	50.25990	50.26270	50.99460
	(!D * RN * !SN * Q)	0.00000	0.00000	0.00000
	(!D * RN * !SN * Q)	16.45320	16.46990	17.30360

TMRDFFSNRNQX1

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Cell Library: Process ,
Voltage 1.80, Temp 25.00

Truth Table

INPUT				OUTPUT
D	RN	SN	CLK	Q
0	1	1	R	0
1	1	1	R	1
x	x	0	x	1
x	0	1	x	0
x	1	1	x	IQ

Footprint

Cell Name	Area
TMRDFFSNRNQX1	774.21521

Pin Capacitance Information

Cell Name	Pin Cap(pf)				Max Cap(pf)
	D	RN	SN	CLK	Q
TMRDFFSNRNQX1	0.04197	0.10235	0.07610	0.07364	5.23714

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
TMRDFFSNRNQX1	0.00000	99.14950	165.06800

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
TMRDFFSNRNQX1	CLK->Q (RR)	0.28656	0.64953	7.25367
	SN->Q (FR)	0.14411	0.57710	8.08002

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
TMRDFFSNRNQX1	CLK->Q (RF)	0.47655	0.79547	5.62365
	RN->Q (FF)	0.36342	0.74581	6.19905
	SN->Q (RF)	0.27525	0.60916	5.75170

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFSNRNQX1	hold	CLK (R)	-0.02025	-0.01600	1.14391
	setup	CLK (R)	0.16883	0.19078	0.66103

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFSNRNQX1	hold	CLK (R)	-0.06911	-0.12802	-1.05399
	setup	CLK (R)	0.10011	0.16329	1.79899

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNRNQX1	hold	CLK (R)	(RN * SN)	-0.02025	-0.01600	1.14391
	setup	CLK (R)	(RN * SN)	0.16883	0.19078	0.66103

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNRNQX1	hold	CLK (R)	(RN * SN)	-0.06911	-0.12802	-1.05399
	setup	CLK (R)	(RN * SN)	0.10011	0.16329	1.79899

Constraints(ns) for RN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFSNRNQX1	recovery	CLK (R)	0.16081	0.19876	6.86475
	removal	CLK (R)	-0.01918	-0.02500	-0.06887
	hold	SN (R)	0.04238	0.07751	0.33066
	setup	SN (R)	0.00701	0.02400	1.34048

Constraints(ns) for RN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNRNQX1	recovery	CLK (R)	(D * SN)	0.16081	0.19876	6.86475
	removal	CLK (R)	(D * SN)	-0.01918	-0.02500	-0.06887
	hold	SN (R)	(CLK * D)	0.00634	-0.01250	-0.11123
	hold	SN (R)	(CLK * !D)	0.00634	-0.01250	-0.11778
	hold	SN (R)	(!CLK * D)	0.04238	0.07751	0.31070
	hold	SN (R)	(!CLK * !D)	0.04238	0.07751	0.33066
	setup	SN (R)	(CLK * D)	0.00367	0.02000	0.52830
	setup	SN (R)	(CLK * !D)	0.00595	0.02400	0.60141
	setup	SN (R)	(!CLK * D)	0.00643	-0.00518	1.34048
	setup	SN (R)	(!CLK * !D)	0.00701	-0.00617	1.06543

Constraints(ns) for RN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNRNQX1	min_pulse_width	RN ()	(CLK * D * SN)	0.12337	0.47852	16.50020
	min_pulse_width	RN ()	(CLK * !D * SN)	0.11612	0.47852	16.50020
	min_pulse_width	RN ()	(!CLK * D * SN)	0.09438	0.47852	16.50020
	min_pulse_width	RN ()	(!CLK * !D * SN)	0.09438	0.47852	16.50020

Constraints(ns) for SN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFSNRNQX1	recovery	CLK (R)	0.01482	0.00822	3.72319
	removal	CLK (R)	0.00266	0.01000	0.01915
	hold	RN (R)	-0.00234	0.01500	0.11574
	setup	RN (R)	0.02695	0.03601	0.31271

Constraints(ns) for SN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNRNQX1	recovery	CLK (R)	(!D * RN)	0.01482	0.00822	3.72319
	removal	CLK (R)	(!D * RN)	0.00266	0.01000	0.01915
	hold	RN (R)	CLK	-0.00234	0.01500	0.11574
	hold	RN (R)	(!CLK * D)	-0.00930	-0.02500	-0.06563
	hold	RN (R)	(!CLK * !D)	-0.01079	-0.02750	-0.07146
	setup	RN (R)	CLK	0.02695	0.01926	0.03046
	setup	RN (R)	(!CLK * D)	0.01811	0.03601	0.31271
	setup	RN (R)	(!CLK * !D)	0.02082	0.03601	0.16182

Constraints(ns) for SN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNRNQX1	min_pulse_width	SN ()	(CLK * D * RN)	0.09800	0.47852	16.50020
	min_pulse_width	SN ()	(CLK * !D * RN)	0.09800	0.47852	16.50020
	min_pulse_width	SN ()	(!CLK * D * RN)	0.11249	0.47852	16.50020
	min_pulse_width	SN ()	(!CLK * !D * RN)	0.11249	0.47852	16.50020

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNRNQX1	min_pulse_width	CLK ()	(D * RN * SN)	0.20672	0.47852	16.50020
	min_pulse_width	CLK ()	(!D * RN * SN)	0.15961	0.47852	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNRNQX1	min_pulse_width	CLK ()	(D * RN * SN)	0.23571	0.47852	16.50020
	min_pulse_width	CLK ()	(!D * RN * SN)	0.11249	0.47852	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
TMRDFFSNRNQX1	CLK	0.00000	0.00000	0.00000
	CLK	97.29270	97.29180	97.86550
	SN	96.81610	96.83830	97.82640

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
TMRDFFSNRNQX1	CLK	0.00000	0.00000	0.00000
	CLK	98.10000	98.10530	98.71070
	RN	-0.00228	-0.31622	-8.48416
	RN	99.74110	99.76340	100.78500
	SN	98.73910	98.74510	99.84780

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNRNQX1	(CLK * RN * SN * !Q) + (CLK * !RN * SN * !Q) + (CLK * !RN * !SN * Q) + (!CLK * !RN * SN * !Q) + (!CLK * !RN * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * RN * SN * !Q) + (CLK * !RN * SN * !Q) + (CLK * !RN * !SN * Q) + (!CLK * !RN * SN * !Q) + (!CLK * !RN * !SN * Q)	95.19040	95.18940	95.37580
	(CLK * RN * Q) + (!CLK * RN * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * RN * Q) + (!CLK * RN * !SN * Q)	101.59900	101.59900	101.97100
	(!CLK * RN * SN)	0.00000	0.00000	0.00000
	(!CLK * RN * SN)	53.89330	53.89220	54.46390

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNRNQX1	(CLK * RN * SN * !Q) + (CLK * !RN * SN * !Q) + (CLK * !RN * !SN * Q) + (!CLK * !RN * SN * !Q) + (!CLK * !RN * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * RN * SN * !Q) + (CLK * !RN * SN * !Q) + (CLK * !RN * !SN * Q) + (!CLK * !RN * SN * !Q) + (!CLK * !RN * !SN * Q)	95.14760	95.14380	95.14550
	(CLK * RN * Q) + (!CLK * RN * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * RN * Q) + (!CLK * RN * !SN * Q)	69.60880	69.61480	70.02420
	(!CLK * RN * SN)	0.00000	0.00000	0.00000
	(!CLK * RN * SN)	53.95830	53.97250	54.60440

Passive power(pJ) for RN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNRNQX1	(CLK * SN * !Q) + (!CLK * !D * SN * !Q)	0.00000	0.00000	0.00000
	(CLK * SN * !Q) + (!CLK * !D * SN * !Q)	91.11110	91.11550	91.11620
	(CLK * D * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * D * !SN * Q)	98.32410	98.32790	99.42450
	(CLK * !D * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * !D * !SN * Q)	53.68440	53.68280	54.49640
	(!CLK * D * SN * !Q)	0.00000	0.00000	0.00000
	(!CLK * D * SN * !Q)	51.99350	51.99150	52.54790
	(!CLK * D * !SN * Q)	0.00000	0.00000	0.00000
	(!CLK * D * !SN * Q)	47.00950	47.01000	48.04150
	(!CLK * !D * !SN * Q)	0.00000	0.00000	0.00000
	(!CLK * !D * !SN * Q)	13.73860	13.73800	14.24690

Passive power(pJ) for RN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNRNQX1	(CLK * SN * !Q) + (!CLK * !D * SN * !Q)	0.00000	0.00000	0.00000
	(CLK * SN * !Q) + (!CLK * !D * SN * !Q)	98.11190	98.11220	98.44060
	(CLK * D * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * D * !SN * Q)	16.10040	16.13390	17.86230
	(CLK * !D * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * !D * !SN * Q)	19.57920	19.59700	20.75270
	(!CLK * D * SN * !Q)	0.00000	0.00000	0.00000
	(!CLK * D * SN * !Q)	56.12790	56.14070	56.87430
	(!CLK * D * !SN * Q)	0.00000	0.00000	0.00000
	(!CLK * D * !SN * Q)	0.34939	0.38532	1.64303
	(!CLK * !D * !SN * Q)	0.00000	0.00000	0.00000
	(!CLK * !D * !SN * Q)	0.27126	0.28988	0.85532

Passive power(pJ) for SN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNRNQX1	(CLK * RN * Q) + (!CLK * D * RN * Q)	0.00000	0.00000	0.00000
	(CLK * RN * Q) + (!CLK * D * RN * Q)	92.00070	92.02570	92.02800
	(!CLK * !D * RN * Q)	0.00000	0.00000	0.00000
	(!CLK * !D * RN * Q)	53.56070	53.56100	54.13200

Passive power(pJ) for SN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNRNQX1	(CLK * RN * Q) + (!CLK * D * RN * Q)	0.00000	0.00000	0.00000
	(CLK * RN * Q) + (!CLK * D * RN * Q)	95.55590	95.58420	95.58590
	(!CLK * !D * RN * Q)	0.00000	0.00000	0.00000
	(!CLK * !D * RN * Q)	19.40530	19.42070	20.02810

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNRNQX1	(D * RN * Q)	0.00000	0.00000	0.00000
	(D * RN * Q)	97.41050	97.40990	97.97280
	(D * !RN * SN * !Q) + (D * !RN * !SN * Q) + (!D * RN * SN * !Q) + (!D * !RN * SN * !Q) + (!D * !RN * !SN * Q)	0.00000	0.00000	0.00000
	(D * !RN * SN * !Q) + (D * !RN * !SN * Q) + (!D * RN * SN * !Q) + (!D * !RN * SN * !Q) + (!D * !RN * !SN * Q)	97.66860	97.66780	98.24940
	(!D * RN * !SN * Q)	0.00000	0.00000	0.00000
	(!D * RN * !SN * Q)	55.04250	55.05700	55.80830

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNRNQX1	(D * RN * SN * !Q)	0.00000	0.00000	0.00000
	(D * RN * SN * !Q)	51.39570	51.40420	52.02710
	(D * RN * Q)	0.00000	0.00000	0.00000
	(D * RN * Q)	55.40960	55.41420	56.06520
	(D * !RN * SN * !Q) + (D * !RN * !SN * Q) + (!D * RN * SN * !Q) + (!D * !RN * SN * !Q) + (!D * !RN * !SN * Q)	0.00000	0.00000	0.00000
	(D * !RN * SN * !Q) + (D * !RN * !SN * Q) + (!D * RN * SN * !Q) + (!D * !RN * SN * !Q) + (!D * !RN * !SN * Q)	51.71970	51.73300	52.38260
	(!D * RN * SN * Q)	0.00000	0.00000	0.00000
	(!D * RN * SN * Q)	57.10700	57.11780	57.78760
	(!D * RN * !SN * Q)	0.00000	0.00000	0.00000
	(!D * RN * !SN * Q)	18.24790	18.26360	19.08370

VOTER3X1

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Truth Table

INPUT			OUTPUT
A	B	C	Y
0	0	x	0
0	1	0	0
x	1	1	1
1	0	0	0
1	x	1	1
1	1	x	1

Footprint

Cell Name	Area
VOTER3X1	102.35440

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	A	B	C	Y
VOTER3X1	0.02277	0.02175	0.02093	5.80372

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
VOTER3X1	0.00000	4.50010	10.36770

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
VOTER3X1	A->Y (RR)	0.06996	0.42707	7.28853
	B->Y (RR)	0.08215	0.42168	6.93881
	C->Y (RR)	0.07602	0.42103	7.26180

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
VOTER3X1	A->Y (FF)	0.13315	0.49970	6.56680
	B->Y (FF)	0.18448	0.54706	6.58751
	C->Y (FF)	0.14824	0.51823	6.68032

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
VOTER3X1	A	5.10103	5.10445	5.27270
	B	3.96490	3.96549	4.20779
	C	5.09947	5.10358	5.26638

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
VOTER3X1	A	2.09435	2.09668	2.28477
	B	1.01346	1.01452	1.15443
	C	2.10256	2.10515	2.31756

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTER3X1	(B * C * Y)	0.00000	0.00000	0.00000
	(B * C * Y)	0.01848	0.01823	0.01822
	(!B * !C * !Y)	0.00000	0.00000	0.00000
	(!B * !C * !Y)	1.45978	1.45927	1.45928

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTER3X1	(B * C * Y)	0.00000	0.00000	0.00000
	(B * C * Y)	3.26541	3.26554	3.26554
	(!B * !C * !Y)	0.00000	0.00000	0.00000
	(!B * !C * !Y)	2.22108	2.22068	2.22132

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTER3X1	(A * C * Y)	0.00000	0.00000	0.00000
	(A * C * Y)	0.04072	0.03716	0.03626
	(!A * !C * !Y)	0.00000	0.00000	0.00000
	(!A * !C * !Y)	1.45484	1.45399	1.45465

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTER3X1	(A * C * Y)	0.00000	0.00000	0.00000
	(A * C * Y)	5.84274	5.84178	5.84141
	(!A * !C * !Y)	0.00000	0.00000	0.00000
	(!A * !C * !Y)	2.21941	2.21760	2.21783

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTER3X1	(A * B * Y)	0.00000	0.00000	0.00000
	(A * B * Y)	0.01867	0.01866	0.01862
	(!A * !B * !Y)	0.00000	0.00000	0.00000
	(!A * !B * !Y)	1.45751	1.45735	1.45682

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTER3X1	(A * B * Y)	0.00000	0.00000	0.00000
	(A * B * Y)	3.95387	3.95393	3.95414
	(!A * !B * !Y)	0.00000	0.00000	0.00000
	(!A * !B * !Y)	2.21986	2.21913	2.21895

VOTERN3X1

sky130_rhbd_tt_1P8_25C.ccs Cell Library:
Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT			OUTPUT
A	B	C	YN
0	0	x	1
0	1	0	1
x	1	1	0
1	0	0	1
1	x	1	0
1	1	x	0

Footprint

Cell Name	Area
VOTERN3X1	85.12720

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	A	B	C	YN
VOTERN3X1	0.02275	0.02178	0.02070	1.76734

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
VOTERN3X1	0.00000	2.49121	10.79050

Delay Information

Delay(ns) to YN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
VOTERN3X1	A->YN (FR)	0.10858	0.65269	9.64520
	B->YN (FR)	0.12992	0.67448	9.51279
	C->YN (FR)	0.08516	0.63248	9.75140

Delay(ns) to YN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
VOTERN3X1	A->YN (RF)	0.04508	0.36144	5.24977
	B->YN (RF)	0.04180	0.32986	4.92438
	C->YN (RF)	0.04417	0.36373	5.43283

Power Information

Internal switching power(pJ) to YN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
VOTERN3X1	A	0.05302	0.05353	0.10264
	B	0.07234	0.07170	0.09839
	C	0.06090	0.06172	0.11395

Internal switching power(pJ) to YN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
VOTERN3X1	A	4.74039	4.73864	4.76584
	B	4.73996	4.73788	4.75969
	C	3.78403	3.78435	3.80408

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTERN3X1	(B * C * !YN)	0.00000	0.00000	0.00000
	(B * C * !YN)	0.01473	0.01453	0.01452
	(!B * !C * YN)	0.00000	0.00000	0.00000
	(!B * !C * YN)	0.00430	0.00426	0.00422

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTERN3X1	(B * C * !YN)	0.00000	0.00000	0.00000
	(B * C * !YN)	3.52069	3.52060	3.52080
	(!B * !C * YN)	0.00000	0.00000	0.00000
	(!B * !C * YN)	0.02479	0.02482	0.02523

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTERN3X1	(A * C * !YN)	0.00000	0.00000	0.00000
	(A * C * !YN)	0.04296	0.03948	0.03849
	(!A * !C * YN)	0.00000	0.00000	0.00000
	(!A * !C * YN)	-0.00067	-0.00070	-0.00078

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTERN3X1	(A * C * !YN)	0.00000	0.00000	0.00000
	(A * C * !YN)	6.24339	6.24177	6.24285
	(!A * !C * YN)	0.00000	0.00000	0.00000
	(!A * !C * YN)	0.02512	0.02421	0.02381

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTERN3X1	(A * B * !YN)	0.00000	0.00000	0.00000
	(A * B * !YN)	0.01447	0.01445	0.01445
	(!A * !B * YN)	0.00000	0.00000	0.00000
	(!A * !B * YN)	0.00202	0.00195	0.00189

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTERN3X1	(A * B * !YN)	0.00000	0.00000	0.00000
	(A * B * !YN)	4.74574	4.74505	4.74547
	(!A * !B * YN)	0.00000	0.00000	0.00000
	(!A * !B * YN)	0.02477	0.02429	0.02427

XNOR2X1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT		OUTPUT
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

Footprint

Cell Name	Area
XNOR2X1	93.74080

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	A	B	Y
XNOR2X1	0.02170	0.02389	2.40112

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
XNOR2X1	0.00000	9.81259	18.16850

Delay Information

Delay(ns) to Y rising (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
XNOR2X1	A->Y (RR)	B	0.06301	0.43227	6.27487
	A->Y (FR)	!B	0.05293	0.57051	9.85113
	B->Y (RR)	A	0.08241	0.47171	6.54265
	B->Y (FR)	!A	0.04251	0.55972	9.74964

Delay(ns) to Y falling (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
XNOR2X1	A->Y (FF)	B	0.06386	0.35289	4.53119
	A->Y (RF)	!B	0.03704	0.36825	6.29072
	B->Y (FF)	A	0.07695	0.36659	4.51702
	B->Y (RF)	!A	0.03203	0.35947	6.29585

Power Information

Internal switching power(pJ) to Y rising (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
XNOR2X1	A	B	0.00000	0.00000	0.00000
	A	B	4.99856	5.00936	5.27697
	A	!B	0.00000	0.00000	0.00000
	A	!B	0.07178	0.08426	0.36493
	B	A	0.00000	0.00000	0.00000
	B	A	1.83025	1.83587	2.05399
	B	!A	0.00000	0.00000	0.00000
	B	!A	0.06853	0.08245	0.40166

Internal switching power(pJ) to Y falling (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
XNOR2X1	A	B	0.00000	0.00000	0.00000
	A	B	11.02840	11.03890	11.30110
	A	!B	0.00000	0.00000	0.00000
	A	!B	8.94338	8.94999	9.18685
	B	A	0.00000	0.00000	0.00000
	B	A	8.98996	8.99851	9.26023
	B	!A	0.00000	0.00000	0.00000
	B	!A	9.34786	9.35449	9.57918

XOR2X1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT		OUTPUT
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Footprint

Cell Name	Area
XOR2X1	93.74080

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	A	B	Y
XOR2X1	0.02186	0.02405	2.39454

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
XOR2X1	0.00000	9.77669	19.21560

Delay Information

Delay(ns) to Y rising (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
XOR2X1	A->Y (RR)	!B	0.06318	0.42887	6.22291
	A->Y (FR)	B	0.05687	0.56897	9.80060
	B->Y (RR)	!A	0.08749	0.44809	6.24335
	B->Y (FR)	A	0.04756	0.55660	9.80022

Delay(ns) to Y falling (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
XOR2X1	A->Y (FF)	!B	0.06361	0.35612	4.58470
	A->Y (RF)	B	0.03263	0.36934	6.30968
	B->Y (FF)	!A	0.07132	0.37278	4.55356
	B->Y (RF)	A	0.02761	0.39223	6.71716

Power Information

Internal switching power(pJ) to Y rising (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
XOR2X1	A	B	0.00000	0.00000	0.00000
	A	B	0.53792	0.54853	0.80542
	A	!B	0.00000	0.00000	0.00000
	A	!B	1.72910	1.74020	2.00977
	B	A	0.00000	0.00000	0.00000
	B	A	0.52828	0.53878	0.79270
	B	!A	0.00000	0.00000	0.00000
	B	!A	1.74475	1.75018	2.01986

Internal switching power(pJ) to Y falling (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
XOR2X1	A	B	0.00000	0.00000	0.00000
	A	B	15.38070	15.39170	15.68710
	A	!B	0.00000	0.00000	0.00000
	A	!B	6.22624	6.23681	6.49188
	B	A	0.00000	0.00000	0.00000
	B	A	15.37930	15.39060	15.68870
	B	!A	0.00000	0.00000	0.00000
	B	!A	6.23157	6.24111	6.46386